

Research article

Study of carbonate alkalinity-induced hepatic tissue damage in Hefang crucian carp (*Carassius auratus*) based on transcriptomic analysisXinyu Lei^{a,1}, Xue Cao^{a,1}, Faye Zhang^b, Qifang Lai^c, Pengcheng Gao^c, Yue-hong Li^{a,*}^a College of Animal Science and Technology, Jilin Agricultural University, Changchun 130118, China^b Georgia Institute of Technology, Atlanta, USA^c East China Sea Fisheries Research Institute, Chinese Academy of Fishery Sciences, China

ARTICLE INFO

Edited by Chris Martyniuk

Keywords:

Carbonate alkalinity stress

Histological analysis

Liver injury

Transcriptomics

ABSTRACT

This study investigated the effects of different sodium bicarbonate (NaHCO_3) concentrations (0 g/L, 1 g/L, and 3 g/L) on Hefang crucian carp (12.0 ± 1.1 g) over a 96-hour period. The experiment is divided into three groups, each with three replicates, and each replicate contains 30 fish. We employed a comprehensive approach integrating histology, physiological and biochemical assays, transcriptomics, as well as artificial intelligence (AI)-assisted analysis. This multifaceted method allowed us to examine changes in gill and liver morphology, osmoregulation, antioxidant capacity, immune response, and physiological metabolism. Results showed that gill and liver tissue damage increased with rising water alkalinity. Serum sodium (Na^+), potassium (K^+), blood ammonia, and gill Na^+/K^+ -ATPase (NKA) levels increased significantly ($p < 0.05$). Hepatic antioxidant enzymes initially increased, then decreased with prolonged stress. Serum and liver immunoenzyme indices were higher in bicarbonate-treated groups compared to controls. Carbonate treatment altered lipid and glucose metabolism in both serum and liver. Transcriptome analysis, enhanced by large language models (LLMs), revealed differentially expressed genes (DEGs) significantly associated with ion binding, transport, apoptosis, and metabolism. In conclusion, excessive carbonate intake in fish alters serum physiological functions and affects hepatic metabolic functions. Crucian carp primarily regulate hepatic antioxidant systems, utilize carbohydrate breakdown for energy requirements, and employ lipids in osmoregulation. This study provides insights into fish adaptation to saline-alkaline environments and offers support for the development of aquaculture in saline-alkaline waters.

1. Introduction

The global demand for aquatic products has increased significantly in recent years, prompting the need to expand aquaculture areas. China possesses approximately 460,000 ha of saline water, representing a significant portion of the world's water resources (Li et al., 2024). However, most of this saline water remains unexploited due to its high carbonate alkalinity, high pH, and ionic imbalance. Recent research has focused on saline and alkaline land improvement technologies, with particular emphasis on the "Treating alkalinity through aquaculture" approach (Nong, 2023).

Saline water typically has a high pH and carbonate (HCO_3^-) alkalinity, which can severely impact fish physiology. High pH inhibits ammonia excretion, increases carbon dioxide (CO_2) excretion, and elevates plasma pH, resulting in respiratory alkalosis. Carbonate alkalinity,

a major stressor for fish, affects their physiological and metabolic functions (Wang et al., 2003). Under these conditions, fish must activate complex regulatory mechanisms to maintain homeostasis, including acid-base and osmotic regulation, nitrogen and respiratory metabolism, and immune system modulation (Perry and Gilmour, 2006).

Recent studies have confirmed various negative effects of saline stress on aquatic animals, including changes in serum immunological indices, oxidative damage due to reactive oxygen species accumulation, and alterations in intestinal flora. Carbonate alkalinity stress specifically causes metabolic alkalosis, impairs ammonia excretion, and affects osmotic pressure in fish (Sun et al., 2018; Fan et al., 2021; Lu et al., 2022). Gills play a crucial role in this process, as they are in direct contact with the external water environment. They contain complex transport epithelial cells that participate in gas exchange, osmotic pressure regulation, acid-base regulation, and nitrogen waste excretion (Evans, 2005;

* Corresponding author.

E-mail address: liyuehong2023@163.com (Y.-h. Li).¹ The authors contributed equally.

Jiang et al., 2019). When exposed to saline water, fish initiate a cascade of responses through their gills, affecting gene expression of functional proteins to enhance stress resistance and adapt to the saline environment (Franklin, 2010). The liver, as the center of lipid synthesis and metabolism in fish, is critical for growth, immunity, and tolerance (Nguyen et al., 2008). Under environmental stress, fish tend to consume more energy to maintain growth and development, leading to changes in liver glycogen, protein, and lipid levels (Tseng and Hwang, 2008; Longwu et al., 2016; Zhao et al., 2016). Additionally, changes in water alkalinity can cause oxidative damage due to the accumulation of reactive oxygen species (ROS) in the liver, potentially decreasing immune defenses and disease resistance in fish (Choi et al., 2008).

This study focuses on Hefang crucian carp, a new variety bred by Academician Liu Shaojun's team at Hunan Normal University. This species possesses advantageous traits such as fast growth rate, palatable meat, strong resistance, and salinity tolerance, making it an ideal subject for studying the physiological mechanisms of salinity tolerance in fish and revealing responses to various environmental stresses (Chun et al., 2020).

As a next-generation high-throughput RNA sequencing technology (RNA-Seq), the transcriptome research method can provide transcriptional information more quickly and accurately, study gene function and structure at the overall level, and reveal the molecular mechanisms of biological regulatory processes when species are subjected to stress (Jin et al., 2024). This method is particularly valuable in revealing the molecular mechanisms of biological regulatory processes when species are subjected to stress. Numerous transcriptome sequencing studies of aquatic animals have been published, providing a theoretical foundation for understanding organismal response mechanisms under various environmental conditions. These studies have expanded our knowledge of gene expression regulation in aquatic animals and revealed their physiological and molecular mechanisms in response to environmental stress and adaptation to different habitats. Examples include studies on the spotted sea bass (*Lateolabrax maculatus*) and marbled flounder (*Pseudopleuronectes yokohamae*), among others (Cui et al., 2019; Yonghang et al., 2023).

Changes in alkalinity pose significant challenges to the sustainable development of aquaculture. Understanding the adverse effects triggered by these changes is crucial for creating a favorable environment for fish growth and development, making it a pressing issue in contemporary aquaculture. Therefore, through studying the effects of carbonate stress at various concentrations on the gill and liver tissue structure, physiological metabolism, and transcriptomics of Hefang crucian carp, this experiment elucidated the stress effects of carbonate on crucian carp at a molecular level. This aims to enhance understanding of the mechanisms underlying fish damage during carbonate exposure, provide theoretical support for subsequent osmotic regulatory mechanisms of Hefang crucian carp, and promote the cultivation in saline and alkaline waters. Furthermore, it aims to advance the development and utilization of saline and alkaline water resources. In addition, this integration of AI-assisted analysis with conventional methods represents a novel approach in aquaculture research, potentially revealing nuanced aspects of fish physiology that might otherwise remain undetected.

2. Materials and methods

2.1. Animal experimental and design

Experimental fish were sourced from the Aquatic Research Base of Jilin Agricultural University. Five hundred robust Hefang crucian carp underwent a two-week acclimation period. Subsequently, 360 fish with an average weight of 12.15 ± 3.56 g were randomly assigned to three experimental groups, each with three replicates of 40 fish. Based on previous research findings (Lei et al., 2024), this study used analytical-grade NaHCO_3 and selected three different carbonate alkalinity gradients: a freshwater control group (Con, 0 g/L), a low-concentration

sodium bicarbonate group (LC, 1 g/L), and a high-concentration sodium bicarbonate group (HC, 3 g/L). The resulting pH values for the Con, LC, and HC groups were measured at 7.5, 9.0, and 9.44, respectively.

During the two-week acclimatization period, water was changed every two days at 50 % volume, with temperature maintained at 23.0 ± 1.0 °C and dissolved oxygen kept above 7.5 mg/L. Fish were fed twice daily (8:00 and 18:00) with feed equivalent to 3 % of their body weight. Following acclimatization, an acute carbonate alkalinity stress test was conducted for 96 h without water changes. Samples were collected at five time points during carbonate exposure: 6 h, 12 h, 24 h, 48 h, and 96 h.

2.2. Sample collection

All animal experiments in this study were conducted in compliance with the ARRIVE (Animal Research: Reporting of In Vivo Experiments) guidelines. The experimental procedures adhered to the following regulatory frameworks:

- The operational guidelines of the Animals (Scientific Procedures) Act 1986
- The EU Directive 2010/63 for the Protection of Animals for Scientific Purposes
- The National Institutes of Health (NIH) Guidelines for the Care and Use of Laboratory Animals

At each sampling point, 10 fish from each replicate group were anesthetized using MS-222 (100 mg/L). Blood samples were collected from the tail vein using disposable 1.5 mL syringes and transferred to 1.5 mL centrifuge tubes. The blood was allowed to clot at 4 °C for 3 h, after which the samples were centrifuged at 3000 r/min for 10 min to obtain serum for analysis.

Gill and liver tissues were collected on ice. A portion of these samples was preserved in paraformaldehyde solution for 48 h for histomorphometric observation. Another portion of liver tissue was rinsed with pre-cooled saline and snap-frozen in liquid nitrogen for transcriptomic analysis. The remaining tissue samples were stored at -80 °C for later homogenization and analysis.

2.3. Biochemical parameter determination

Various biochemical parameters were analyzed according to the manufacturers' instructions. The analysis included:

- Electrolytes and enzymes: sodium (Na^+), potassium (K^+), Na^+/K^+ -ATPase (NKA)
- Nitrogen metabolism: blood ammonia, urea (BUN), glutamine synthetase (GS)
- Liver function: aspartate aminotransferase (AST), alanine aminotransferase (ALT)
- Oxidative stress markers: antioxidant enzymes (SOD, CAT, GSH-Px), malondialdehyde (MDA)
- Immune factors: complement factors (C3, C4), immunoglobulin M (IgM)
- Phosphatases: acid phosphatase (ACP), alkaline phosphatase (AKP)
- Lipid profile: triglycerides (TG), total cholesterol (T-CHO), low-density lipoprotein (LDL-C), high-density lipoprotein (HDL-C)
- Carbohydrate metabolism: glucose (Glu), liver glycogen
- Metabolic enzymes: glutamate dehydrogenase (GDH), acetyl-CoA carboxylase (ACC), carnitine palmitoyltransferase I (CPT-I), acetyl-CoA synthetase (ACS), hepatic lipase (HL), lipoprotein lipase (LPL), hexokinase (HK), phosphofructokinase (PFK), pyruvate kinase (PK), phosphoenolpyruvate carboxykinase (PEPCK).

2.4. Histological analyses

Fixed tissue samples underwent a series of dehydration steps using increasing ethanol concentrations (70 % to 100 %), followed by xylene clearing and paraffin embedding. The samples were then sectioned to 5 μm thickness using a sectioning machine (RM2235, Haonos Technology Co., Ltd., Beijing, China) and stained with hematoxylin and eosin (H&E). Subsequently, histological sections of gills and livers were photographed and observed using an Axioskope microscope (IX 7; Olympus). This allowed for a comparison of changes between the carbonate-treated groups and the control group.

2.5. mRNA library construction and sequencing

Collected samples were sent to Liankawa Biotech for transcriptome sequencing. Total RNA was isolated and purified using Trizol reagent (ThermoFisher, 15596018) following the manufacturer's procedure. Samples with concentrations $>50\text{ ng}/\mu\text{L}$, RIN values >7.0 , and total RNA $>1\text{ }\mu\text{g}$ were selected for downstream experiments. cDNA was synthesized from mRNA carrying polyadenylate using Dynabeads oligo (dT) magnetic beads (Thermo Fisher, USA) and Invitrogen SuperScript™ II Reverse Transcriptase.

The ends of the duplex DNA were aligned into flat ends using *E. coli* DNA polymerase I (NEB, cat.m0209, USA), RNase H (NEB, cat.m0297, USA), and dUTP Solution (Thermo Fisher, cat.R0133, CA, USA), then screened and purified using magnetic beads. Two strands were digested with UDG enzyme (NEB, cat.m0280, MA, US) to form a strand-specific library with a fragment size of $300\text{ bp} \pm 50\text{ bp}$. Sequencing was performed using Illumina Novaseq™ 6000 (LC Bio Technology CO., Ltd. Hangzhou, China) in PE150 mode.

After obtaining the sequencing data, it was filtered to obtain high-quality clean data. This clean data was then compared to the reference genome of the project species, followed by gene expression quantitative analysis, GSEA, gene difference analysis, and enrichment analysis.

2.6. Real-time PCR verification

To confirm the reliability of the transcriptome analysis results, 10 differentially expressed genes (DEGs) with significant changes were randomly selected for qRT-PCR verification. The validation method followed the operation procedure of Zhu (Zhu, 2023), as shown in Table 1.

Table 1
Primers used in this study.

Primers name	Sequence (5'-3')	NCBI-GenelD
β -actin F	GATGATGAAATTGCCGCACTG	AB039726
β -actin R	ACCGACCATGACGCCCTGATGT	
Hb α F	ATTGTGAAGGGCTTGTGGGCTAAG	LOC113049502
Hb α R	GGGAGAAGTAGGCTTGGTCTGAGG	
ERRFI 1 F	TCCGAGGAGTTGTGGACAGATTAGG	LOC113055419
ERRFI 1 R	TTCTGCTTGGCTGTTGGACCTTC	
NF- κ b inhibitor α F	CCACITAGCAGTTGACCTCCAGAAC	LOC113117624
NF- κ b inhibitor α R	ATGTGACCGCCGTATGTGAAGC	
Camk 1 α F	AGCAGCATTAGCAGCGGCAAG	LOC113040340
Camk 1 α R	CGGAGAGTGAAGCGTATTCTG	
glcea F	AGGTCTATCTATGGCATCGGCTCTC	LOC113043709
glcea R	CGGCGTGGCATGATCTTGGTAAC	
Kcnab 2b.F	AGCAGCATTAGCAGCGGCAAG	LOC113094725
Kcnab 2b R	AGACTCCTAATCTCACCCAGTTCCC	
Adgrb 2 F	CTCTTGTGTGGTTCTCCCTCTGTG	LOC113119388
Adgrb 2 R	GTTGATCGGCGGTCTGTGATGG	
Cpxm 2 F	GGTGATTACTGGCGTCTGCTGAAC	LOC113111417
Cpxm 2 R	TGCTGGCTCCGATGTCGTATCC	
Chac 1 F	CCTGTGGATCTTCGGTTACGGTTC	LOC113063007
Chac 1 R	CGTGCCAGAAAGCGTCTCTTGATC	
grik 1a F	TTACGAGGACAGCACAGGATTGATG	LOC113109854
grik 1a R	TTGGCGGATACGAATCTTCAGGTTG	

2.7. Statistical analysis

All enzyme activity data were analyzed by one-way ANOVA using SPSS (23.0). Duncan's multiple comparisons were used when the one-way ANOVA showed significant differences ($p < 0.05$). Data obtained from the experiments were expressed as "mean $\pm$ standard deviation".

2.8. AI-assisted data analysis

To complement traditional bioinformatics approaches, we employed advanced LLMs and AI tools to enhance our analysis. We used GPT-4 (OpenAI) to analyze differentially expressed genes (DEGs) and their functional annotations, identifying complex patterns and relationships between genes. For KEGG pathway analysis and GO enrichment analysis, we utilized BioBERT, a biomedical language representation model pre-trained on large-scale biomedical corpora. BioBERT interpreted KEGG pathway enrichment results, identifying potential cross-talk between pathways and suggesting novel biological mechanisms. It also provided deeper insights into enriched GO terms, identifying overarching themes and potential interactions between different GO categories. All AI-generated insights were critically evaluated for biological plausibility and consistency with experimental data before incorporation into our analysis. The integration of these advanced AI tools allowed us to uncover complex patterns and relationships in our data that might not have been apparent through conventional analysis methods alone.

3. Result

3.1. Histological analyses

Changes in the histomorphology of gill and liver tissues of crucian carp in each group after 96 h of carbonate stress were recorded. The gill tissues of the Con group were structurally intact without obvious pathological changes. Gill lamellae were comb-like extended, neatly arranged, and almost parallel, with regular distribution of column cells. Compared to the Con group, the gill lamellae of the LC group were slightly curved, while those of the HC group were more curved, almost in an "S" shape (Fig. 1).

In the carbonate-treated groups, gill filaments of Hefang crucian carp became narrower, the ends of gill lamellae were enlarged, and the number of interlamellar cell mass (ILCM) cells was significantly reduced. The lamellar thickness of gill lamellae ($2.87 \pm 0.42\text{ }\mu\text{m}$ and $3.23 \pm 0.54\text{ }\mu\text{m}$, respectively) was smaller than that of the control group ($3.25 \pm 0.47\text{ }\mu\text{m}$), with a significant difference in the LC group ($p < 0.05$). The height of gill lamellae was positively correlated with carbonate concentration, with significant differences between the HC group and the LC and Con groups ($p < 0.05$). The lamellar base length showed a positive correlation with increasing carbonate concentration, initially decreasing and then increasing, but differences among the three groups were not significant ($p > 0.05$). Specific changes are shown in Table 2.

In the Con group, hepatocytes were closely arranged in spheres or polygons of different sizes, with most nuclei located in the middle of the hepatocytes and clear outlines of hepatic lobules. In the LC group, some liver nuclei showed nuclear migration and nucleolysis, blurred outlines of liver lobules, cellular vacuolation, and a few sinusoid channels appeared, indicating slight damage to the liver tissue structure. In the HC group, cell arrangement was disordered, outlines of some liver lobules disappeared, numerous liver nuclei scattered like stars and migrated around the cells, and the number and width of sinusoid channels increased compared to the LC group (Fig. 2).

3.2. Changes in osmotic regulation

Fig. 3 illustrates the osmoregulatory changes in test fish after 96 h of exposure to different carbonate concentrations. Levels of Na^+ , K^+ , and

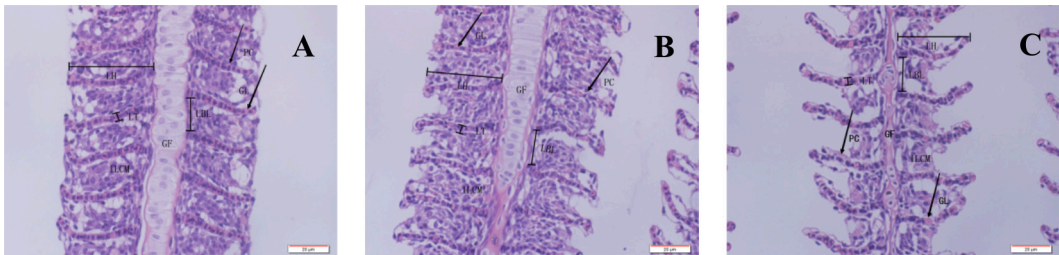


Fig. 1. Histomorphological changes of gills of Hefang crucian carp after exposure to carbonate alkalinity for 96 h (400×). LT-sheet thickness; LBL-lamellar base length; LH-layer height; GF-gill filament; GL-branchial lamella; PC-column cells. H&E, scale = 20 μm.

Table 2
Changes of gill lamella of Hefang crucian carp under different carbonate alkalinity (μm).

Alkalinity (g/L)	LT	LH	LBL
0.0	3.25 ± 0.47a	35.48 ± 5.74b	17.80 ± 2.74a
1.0	2.87 ± 0.42b	37.41 ± 2.55b	17.26 ± 1.94a
3.0	3.23 ± 0.54a	40.67 ± 4.53a	18.48 ± 2.47a

Different letters in the same column show significant differences ($p < 0.05$). LT: Lamellar thickness; LH: Lamellar height; LBL: Lamellar basal length.

NKA in the carbonate-treated groups increased with rising carbonate concentration compared to the Con group. Serum analysis after 96 h of carbonate stress revealed significantly higher Na^+ concentrations in the carbonate-treated groups compared to the Con group, except at 6 h and 12 h ($p < 0.05$). Serum K^+ concentrations were significantly higher in the stress groups at 12 h and 24 h compared to the Con group ($p < 0.05$). Gill NKA activity was significantly elevated in all three groups within 96 h of carbonate stress ($p < 0.05$). All groups exhibited a similar trend, initially increasing and then decreasing with stress time, peaking at 24 h.

3.3. Changes of ammonia transport indexes

The changes of ammonia transport related indexes in the serum of Hefang crucian carp exposed to different concentrations of carbonate for 96 h are shown in Fig. 4. Compared with the Con group, blood ammonia levels in the carbonate-treated group at the five time points increased significantly ($p < 0.05$) with the increase in carbonate concentration, but at 24 h of stress, the difference between the carbonate-stressed groups was not significant ($p > 0.05$). With the increase of stress time, blood ammonia concentration in Con, LC and HC groups showed a trend of increasing and then decreasing, reaching the maximum value at 24 h. Compared with the Con group, the BUN levels in the carbonate stress group showed a decreasing trend with the increase of carbonate concentration, and the differences among the three groups were not significant ($p > 0.05$).

After 96 h of acute carbonate alkalinity stress, GS concentrations in gills and liver were higher in the carbonate-stressed groups than in the

Con group. GS activity in gills of carbonate-treated groups was significantly increased ($p < 0.05$) compared to the Con group, except at 6 h and 96 h. Liver GS activity in carbonate-treated groups was significantly higher ($p < 0.05$) than the Con group only at 12 h, with both gill and liver GS activity peaking at 24 h. GDH activity in gills and liver reached maximum levels at 24 h in all groups. Gill GDH activity was significantly higher in the HC group compared to LC and Con groups ($p < 0.05$), except at 12 h and 24 h. In the liver, GDH concentration differences were not significant ($p > 0.05$) at 12 h and 96 h between HC and other groups. Notably, liver GDH concentration was about twice that in the gills.

3.4. Antioxidant ability

Fig. 5 shows the effects of acute carbonate stress on antioxidant indices of Hefang crucian carp. Serum AST and ALT levels increased with carbonate alkalinity concentration and showed an increasing then decreasing trend over time. AST activities in the HC group were significantly higher than in the Con group throughout the 96 h stress period ($p < 0.05$). LC group AST activity didn't differ significantly from the Con group except at 48 h ($p > 0.05$). ALT activities were higher in all carbonate-treated groups than in the Con group, with significant differences between groups ($p < 0.05$).

Activities of antioxidant enzymes (SOD, CAT, and GSH-Px) in all

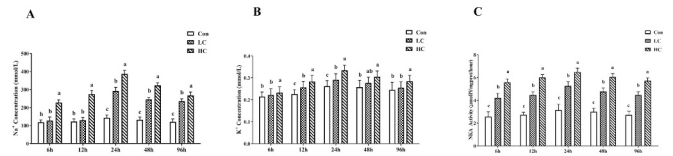


Fig. 3. Effects of different concentrations of carbonate alkali exposure on the participation of serum and gill tissue in osmoregulation of Hefang crucian carp. Serum sodium concentration (Na^+); (B) Serum potassium concentration (K^+); (C) Na^+/K^+ -ATPase activity (NKA). Different capital letters indicated that there was significant difference between different time points in the same concentration group ($p < 0.05$). Different lowercase letters indicate a significant difference between different concentration groups at the same time point ($p < 0.05$), the following figure shows the same.

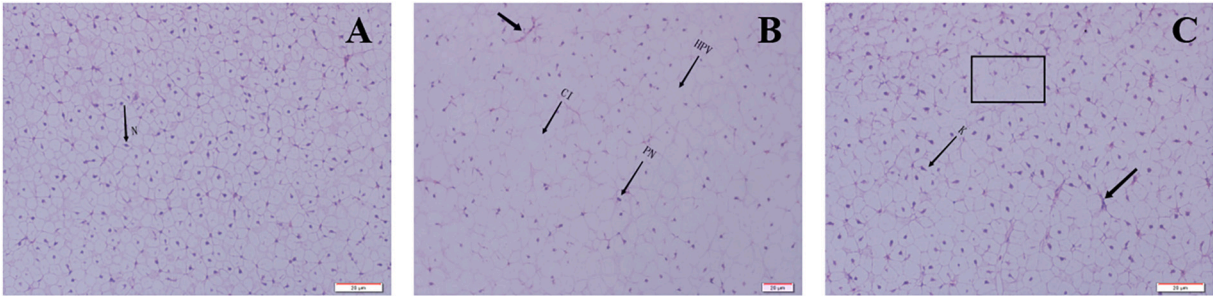


Fig. 2. Liver histomorphologic changes of Hefang crucian carp after 96 h of carbonate alkalinity stress (200×). N-nucleus; HPV-hepatocyte vacuolation; CI-the outline of hepatic lobule is blurred; K-nucleolysis; PN, nuclear migration; Thick arrow-sinus channel; Box-Liver lobular outline disappeared. H&E, scale = 20 μm.

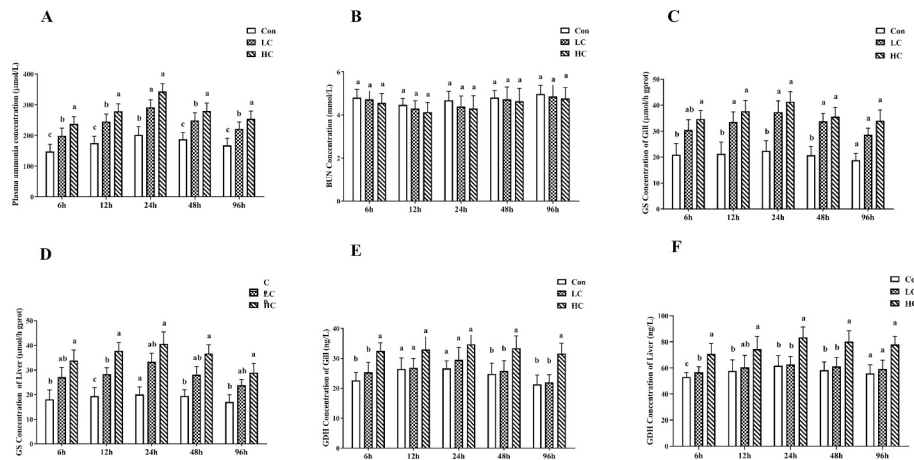


Fig. 4. Effects of different concentrations of carbonate alkalinity exposure on ammonia metabolism in serum, liver and gill tissue of Hefang crucian carp. (A) Blood ammonia; (B) Urea nitrogen (BUN); (C) Glutamine synthetase (GS); (D) Glutamine synthetase (GS); (E) Glutamate dehydrogenase (GDH); (F) Glutamate dehydrogenase (GDH).

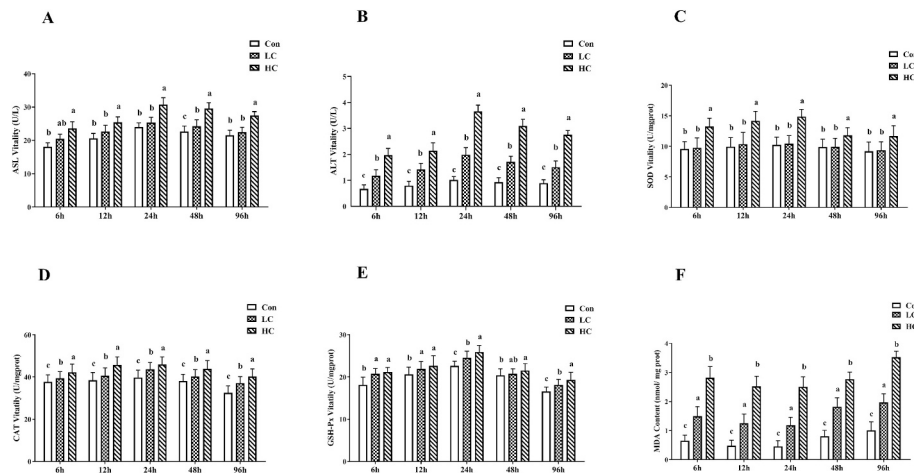


Fig. 5. Changes of antioxidant enzymes in Hefang crucian carp exposed to different concentrations of carbonate alkalinity. (A) Glutamic oxaloacetic transaminase (ASL); (B) Alanine aminotransferase (ALT); (C) Superoxide dismutase (SOD); (D) Catalase (CAT); (E) Glutathione peroxidase (GSH-Px); (F) Malondialdehyde (MDA).

groups gradually increased during the first 24 h of acute stress, peaking at 24 h, then gradually decreased to levels similar to those at 6 h. MDA activity showed the opposite trend. Hepatic SOD activity was significantly elevated only in the HC group ($p < 0.05$). CAT and MDA activities were significantly elevated among all groups within the 96 h stress period ($p < 0.05$). GSH-Px activity in the HC group didn't differ significantly from the LC group at 6 h, 12 h, and 48 h ($p > 0.05$), but both carbonate-treated groups showed significantly higher activity than the Con group except at 48 h ($p < 0.05$).

3.5. Immune response

As shown in Fig. 6, complement C3, complement C4, and IgM levels exhibited a decreasing then increasing trend with stress time at the same carbonate concentration. After 96 h of carbonate alkalinity stress, complement C3 activities in both LC and HC groups were significantly higher than in the Con group ($p < 0.05$). No significant changes in C3 activity were observed between carbonate treatment groups at 12 h and 24 h ($p > 0.05$). Serum complement C4 activity increased with carbonate concentration, but differences were not significant between HC, LC, and Con groups ($p > 0.05$). Blood IgM activity was elevated in carbonate-treated groups compared to the Con group, but differences between

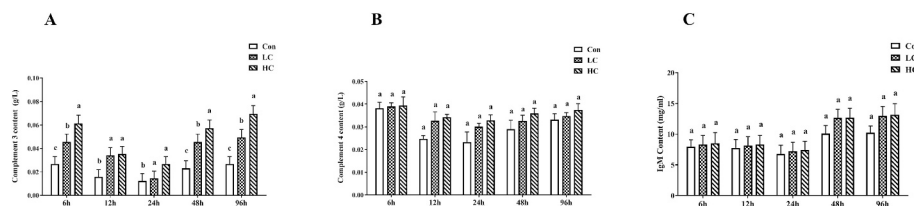


Fig. 6. Effects of different concentrations of carbonate alkalinity on serum immunoenzymes of Hefang crucian carp. (A) Immunoglobulin (IgM); (B) Complement C3 (C3); (C) Complement C4 (C4).

groups were not significant ($p > 0.05$).

Fig. 7 shows liver immune index results. ACP activity was significantly elevated in the carbonate stress groups compared to the Con group ($p < 0.05$), except at 6 h and 12 h. AKP activity in the liver increased significantly during the stress period in all cases ($p < 0.05$). Enzyme activities increased significantly at the beginning of the stress period, peaked at 24 h, and then gradually decreased.

3.6. Effects of carbonate alkaline stress on liver metabolism

3.6.1. Lipid metabolism

Fig. 8 presents results of lipid metabolizing enzymes in serum. Within 96 h of acute stress, TG, T-CHO, LDL, and HDL contents in the carbonate stress groups were lower than in the Con group, showing an increase then decrease with stress time. TG content differences were significantly lower in all groups except at 12 h ($p < 0.05$). At 96 h, only the Con group had significantly higher T-CHO content compared to carbonate stress groups ($p < 0.05$). LDL content showed a negative correlation with carbonate concentration, being significantly lower in carbonate-treated groups than in the Con group, except at 48 h ($p < 0.05$). No significant changes in LDL content were observed between HC and LC groups at 6 h, 24 h, 48 h, and 96 h ($p > 0.05$). Serum HDL content changes were not significantly different within 96 h of stress ($p > 0.05$).

Fig. 9 illustrates the effects of acute carbonate stress on hepatic lipid metabolism indices. Liver lipid metabolizing enzyme levels were higher in carbonate-treated groups than in the Con group and increased with carbonate concentration. HL and LPL activities differed significantly between the three groups ($p < 0.05$), while ACS activity differences were not significant in the LC group compared to the Con group ($p > 0.05$). FAS content showed fluctuating changes with stress time, being significantly higher in the HC group compared to the Con group except at 24 h and 96 h ($p < 0.05$). CPT-1 levels decreased then increased with increasing stress time.

3.6.2. Carbohydrate metabolism

Fig. 10 illustrates the Glu content in the serum of Hefang crucian carp under different alkalinity conditions. Serum Glu content peaked at 24 h and showed a decreasing trend afterward. Within 96 h of acute carbonate stress, the Glu content in the HC group was significantly higher than in the LC and Con groups ($p < 0.05$).

Fig. 11 depicts the effects of acute carbonate stress on hepatic glucose metabolism indices. After 96 h of acute carbonate stress, the HK content in the liver of the carbonate-stressed group was significantly lower compared to the control group ($p < 0.05$). PFK and PK changes showed consistent trends, both increasing and then decreasing in response to rising carbonate concentration. PEPCK content increased with carbonate concentration across groups. The difference between the LC group and the Con and HC groups was not significant ($p > 0.05$), while PEPCK content in the HC group was significantly higher than in the Con group ($p < 0.05$).

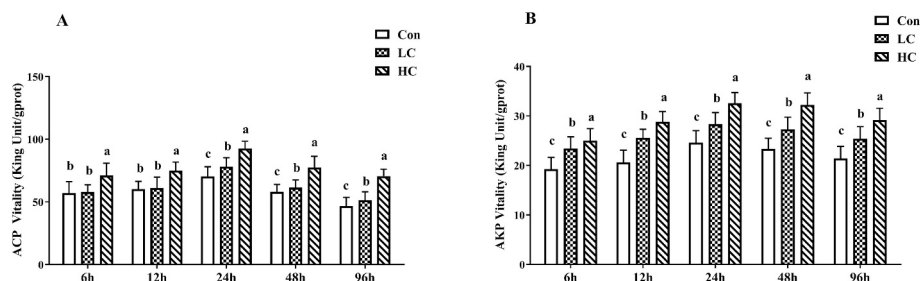


Fig. 7. Effects of different concentrations of carbonate alkalinity on liver immunoenzymes of Hefang crucian carp. (A) Acid phosphatase (ACP); (B) Alkaline phosphatase (AKP).

3.7. Analysis of liver transcriptome under carbonate alkaline stress

3.7.1. Data quality control analysis

RNA-seq was performed on liver samples collected after 96 h of exposure to different carbonate alkaline concentrations (Con group; HC group), yielding 220.55 million raw reads. Filtering resulted in 215.41 million high-quality clean reads. The Q20 percentage reached 99.98 %, and the Q30 percentage exceeded 96.60 % in both groups. Detailed sequencing statistics are presented in Table 3.

3.7.2. Differentially expressed genes

Comparison of the 96 h HC group and Con group identified 3394 differentially expressed genes (DEGs) (adjusted $p < 0.05$ and $|\log_2 \text{Fold Change}| \geq 1$). Fig. 12 presents a histogram of DEGs, showing 1145 up-regulated and 2249 down-regulated genes compared to the Con group. A volcano plot visually represents the DEG distribution, with up-regulated genes in red, down-regulated in blue, and non-significant genes in gray.

3.7.3. GO enrichment analysis

GO enrichment annotation of DEGs, shown in Fig. 13, categorized genes into Biological Processes (BP), Molecular Function (MF), and Cellular Component (CC). BP included 25 pathways, primarily enriched in Biological Process (237 DEGs), Transcriptional Regulation (177 DEGs), and Signal Transduction (132 DEGs), with significant enrichment in lipid metabolism. CC analysis revealed high numbers of differentially significant genes in Membrane (916 DEGs) and Membrane Integral Component (885 DEGs). MF analysis showed numerous differentially significant genes in Metal Ion Binding (434 DEGs), ATP binding (273 DEGs), and nucleotide binding (213 DEGs).

Fig. 14 presents an enrichment factor plot of the top 20 GO terms with the smallest P -values. The analysis reveals a concentration of DEGs related to metal ion binding, ATP binding, membrane function, and insulin secretion regulation. Appendix 1 lists some of the differentially expressed genes involved in material transport.

3.7.4. KEGG enrichment analysis

KEGG functional enrichment analysis annotated 116 signaling pathways, including cellular processes (17; 14.6 %), environmental information processing (17; 14.6 %), genetic information processing (10; 8.6 %), human diseases (8; 6.8 %), metabolism (41; 35.3 %), and biological systems (23; 19.8 %). Fig. 15 displays a histogram of the top 5 KEGG products by significance. The top five pathways were cell molecular adhesion, MAPK signaling, phagosome, cell senescence, and apoptosis. Cellular molecular adhesion emerged as the main signaling pathway, followed by MAPK signaling and phagosome. High carbonate stress also induced iron death and altered pathways associated with simplex sporadic virus infection. Metabolic signaling pathways were dominated by arachidonic acid metabolism.

3.7.5. Validation of RNA-Seq data

To verify RNA-Seq data accuracy, 10 differentially expressed genes with high enrichment were randomly selected for qRT-PCR verification.

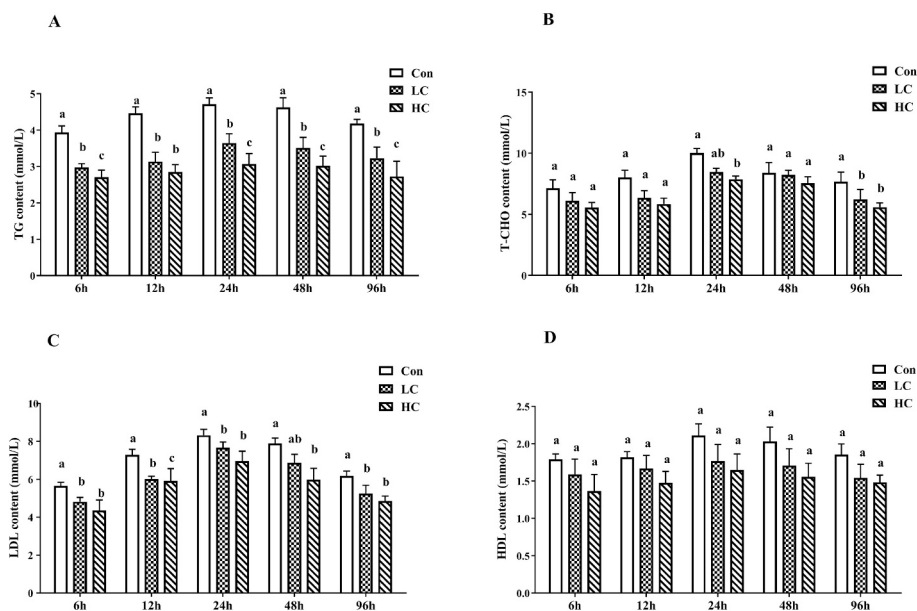


Fig. 8. Effects of different concentrations of carbonate alkalinity on serum lipid metabolism of Hefang crucian carp. (A) Triglycerides (TG); (B) Total cholesterol (T-CHO); (C) Low-density lipoprotein (LDL); (D) High-density lipoprotein (HDL).

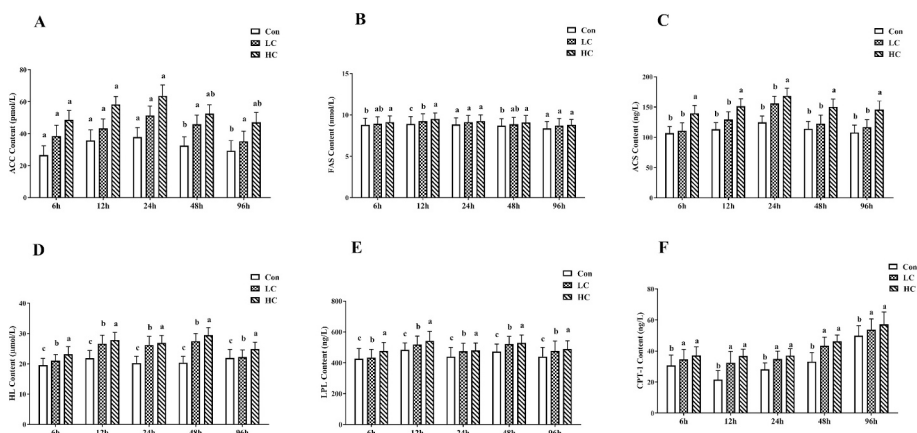


Fig. 9. Effects of different concentrations of carbonate alkalinity on liver lipid metabolism of Hefang crucian carp. (A) Acetyl-CoA carboxylase (ACC); (B) Fatty acid synthase (FAS); (C) Acetyl-CoA synthetase (ACS); (D) Liver lipase (HL); (E) Lipoprotein lipase (LPL); (F) Carnitine palmitoyl transaminase-1 (CPT-1).

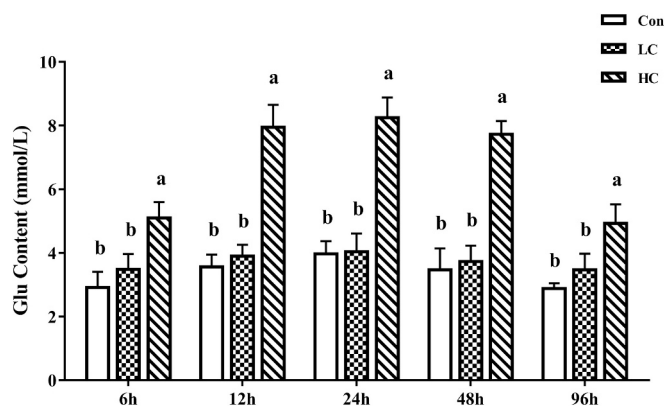


Fig. 10. Effects of different concentrations of carbonate alkalinity on serum glucose content of Hefang crucian carp.

Fig. 16 shows that the relative expression levels of these genes closely match the RNA-Seq analysis results, confirming the reliability of the transcriptome data. The raw sequence data have been submitted to the NCBI Gene Expression Omnibus (GEO) datasets with accession number GSE279585.

4. Discussion

The ability of aquatic animals to adapt to salinity has long been a subject of great interest in aquatic biology. Previous studies have demonstrated that increased salinity in the water column can induce a range of adverse effects in fish, including respiratory and metabolic disturbances, excessive mucus secretion, hemorrhage, and even death (Kultz, 2015). Furthermore, while salinity affects fish growth and development, alkalinity has been shown to inhibit ammonia excretion in fish (Schreck et al., 2011). Changes in alkalinity can lead to “alkaline deposition disease” in aquatic animals and cause various physiological and biochemical abnormalities.

When fish are exposed to carbonate alkalinity, firstly, it will affect the osmotic pressure balance of the fish through the gills, change the ion

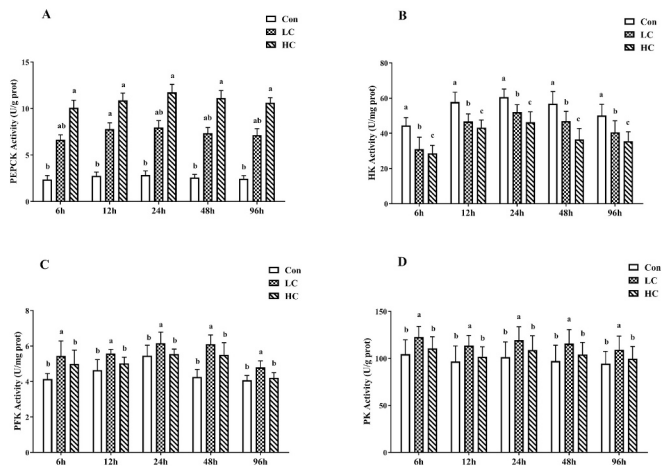


Fig. 11. Effects of different concentrations of carbonate alkalinity on liver glucose metabolism of Hefang crucian carp. (A) Phosphoenolpyruvate kinase (PEPCK); (B) Hexokinase (HK); (C) Phosphofructokinase (PFK); (D) pyruvate kinase (PK).

concentration in the blood, and destroy the ion exchange system of the gill tissues, which will cause physiological and metabolic dysfunctions in the body; on the other hand, the ammoniacal nitrogen (NH_3) cannot be discharged out of the body in the form of ionic ammonia (NH_4^+), and ultimately lead to ammonia toxicity of the fish, which is one of the important causes of fish death in high alkalinity environments (Peh et al., 2010; Fang et al., 2017). Therefore, fish can maintain osmotic pressure homeostasis by increasing mitochondria-rich cells in the gills, controlling plasma ion concentrations, and regulating the levels of ion-transporting enzymes, hormones, and amino acids (Mylonas et al., 2009). Among them, gill NKA enzymes act as the main transporter

enzymes to maintain plasma ion homeostasis by pumping Na^+ and K^+ across the plasma membrane in order to maintain the ion permeability of the cell membrane. The NKA enzyme activity in the gills of most extensively saline scleractinian fishes undergoes adaptive changes in response to changes in alkalinity to meet the gill requirements for internal and external ion transport (Wang et al., 2014a). K^+ significantly influences the transmembrane potential, and in hypertonic environments, gills become more permeable to K^+ , resulting in greater efflux than influx (Listiyani et al., 2018). The present study's results align with those of Liu et al., who observed an increasing trend in serum K^+ concentration of naked carp (*Gymnocypris przewalskii*) in saline water (Liu et al., 2012). Whereas, when medaka (*Oryzias latipes*) was transferred from freshwater to alkaline water environment with pH 9.2, K^+ concentration in plasma decreased significantly ($p < 0.05$); Na^+ concentration increased from 12 h after transfer until the end of the experiment; at the same time, the expression of NKA- α in gills in the carbonate-treated group was significantly increased by 8-fold ($p < 0.05$) (Lin et al., 2015). The differences in the above studies indicate that the cell membranes have different degrees of integrity in different alkalinity ranges, but all have the ability to actively regulate osmotic pressure, so leading to different adaptive abilities of the fish to alkaline conditions. This shows that Hefang crucian carp has the ability to regulate osmotic pressure and adapt to changes in alkaline levels in the water body. In order to adapt to the alkaline water environment, the gill ion transporter enzyme activity of Hefang crucian carp was elevated, the ion efflux mechanism was activated, and the serum ion concentration gradually decreased. The serum ion concentration and ion transporter enzyme activity peaked at 24 h under each concentration gradient in the experiment, and it was hypothesized that 24 h might be the critical point of osmotic pressure regulation in Hefang crucian carp.

As carbonate alkalinity increases, exogenous ammonia nitrogen penetrates the bloodstream through gills and epithelial cells on the body surface. The inhibition of excess ammonia nitrogen accumulation and excretion affects blood ammonia levels, potentially leading to ammonia

Table 3
Statistics for Illumina short reads of the liver transcriptome.

Sample	Raw date read	Base	Valid date read	Base	Valid ratio (reads)	Q20 (%)	Q30 (%)	GC content (%)
Con_1	38,372,798	5.76G	37,533,946	5.63G	97.81	99.98	97.52	45.50
Con_2	43,150,280	6.47G	42,063,044	6.31G	97.48	99.98	96.60	46.50
Con_3	31,491,086	4.72G	30,751,532	4.61G	97.65	99.98	97.48	45.00
HC_1	36,871,770	5.53G	36,077,152	5.41G	97.84	99.98	97.55	45.50
HC_2	35,254,418	5.29G	34,467,250	5.17G	97.77	99.98	97.49	45.50
HC_3	35,412,728	5.31G	34,513,812	5.18G	97.46	99.98	97.49	45.50

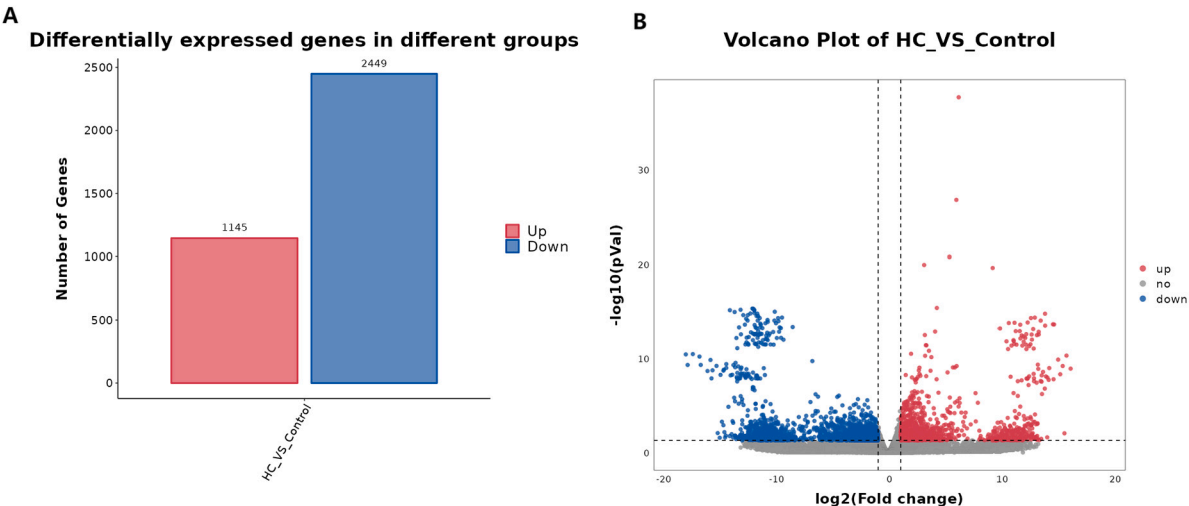


Fig. 12. Transcriptome differential gene expression. (A) Histogram of differentially expressed genes; (B) Volcano Plot.

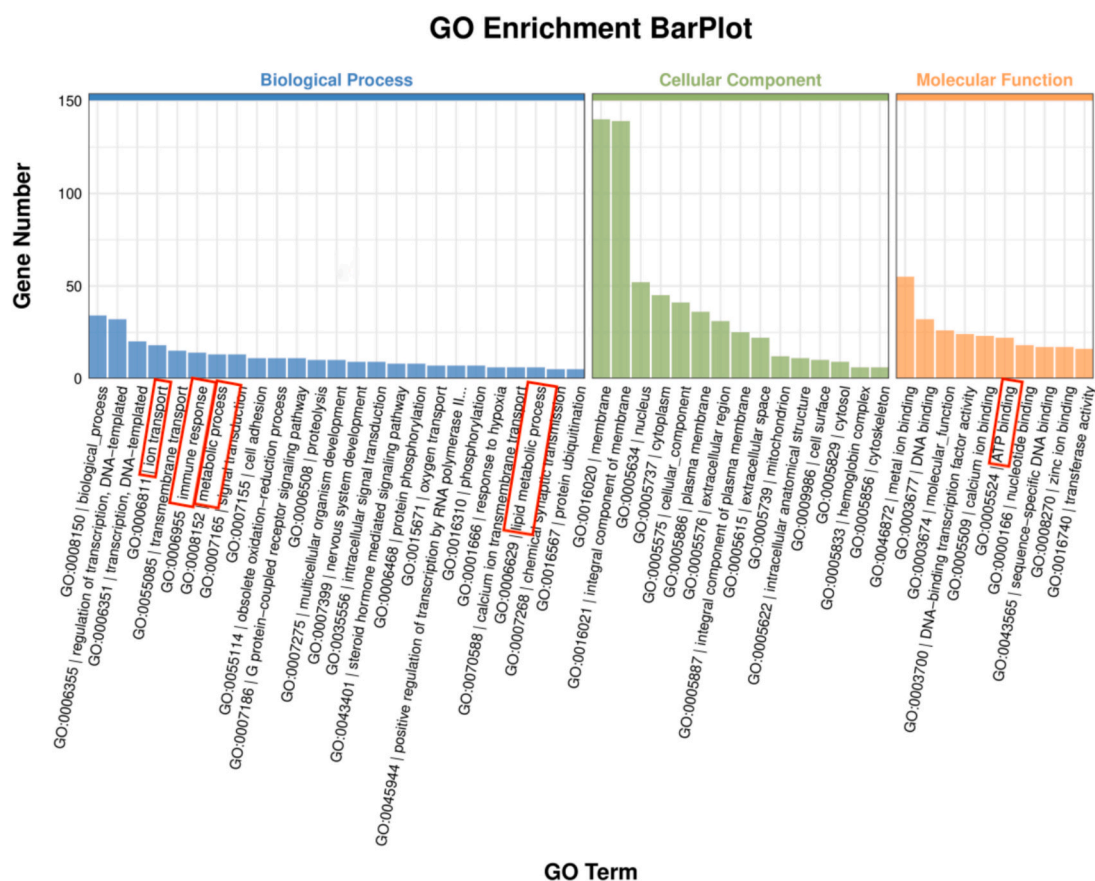


Fig. 13. GO enrichment functional classification.

toxicity. Consequently, ammonia excretion is a crucial physiological function in fish saline and alkaline metabolism (Guo et al., 2014). To cope with the challenge of ammonia excretion under alkaline stress, fish can reduce the ammonia levels in the body by activating the urea and glutamine metabolism pathways. The ammonia in the fish bloodstream is catalyzed by a series of biochemical reactions to compounds such as carbamyl phosphate by carbamyl phosphate synthetase (CPS), finally producing urea that is excreted through the kidney and the gills along with urea transport proteins (Wilkie, 1997; Li et al., 2021). Studies have shown that ammonia stress increases the levels of blood ammonia and urea within the serum, demonstrating that the urea metabolism pathway of ammonia has begun to function in fish at this time (Feng et al., 2023). In the present study, although the blood ammonia content was positively and significantly correlated with the increase in carbonate concentration, there were no significant changes in serum urea levels were observed under different conditions of alkalinity stress, indicating that urea metabolism was not the main pathway for ammonia reduction in Hefang crucian carp. Similar findings have been reported in Chinese black sleeper (*Bostrichthys sinensis*) and gilthead seabream (*Sparus aurata*) (Anderson et al., 2002; Alves et al., 2010).

Increasing glutamine synthetase (GS) levels to synthesize the non-toxic intermediate glutamine is a crucial strategy for ammonia detoxification in fish. After synthesis, glutamine is released from tissues as a non-toxic ammonia metabolite into the bloodstream and ultimately to the liver (Gao et al., 2017).

Under chronic ammonia exposure, GS activity in the brain of Nile tilapia (*Oreochromis niloticus*) showed a significant 2–3 fold increase, driving amino acid synthesis and accumulation (Hegazi et al., 2010). Acute ammonia exposure has been shown to significantly increase the activities of GS and glutamate dehydrogenase (GDH) in the gills and liver of cuttlefish (*Sepia pharaonis*), with both enzymes responding

similarly to ammonia exposure. This demonstrates a simultaneous increase in ammonia detoxification levels in both gills and liver (Peng et al., 2017).

In tiger squid, GS and GDH activities in gills and liver, as well as BUN in hemolymph, are significantly elevated in a concentration-dependent manner with ammonia nitrogen exposure (Peng et al., 2017). While squid can detoxify ammonia to glutamine or urea under high ammonia conditions or impaired ammonia excretion, this ability has limitations. The present study observed similar changes in GS and GDH activities in gills and liver, suggesting that ammonia detoxification levels in both organs were elevated simultaneously during carbonate alkalinity stress. This indicates a similar regulatory mechanism for ammonia detoxification in both tissues.

Notably, the amount of GDH in the liver was about 2–3 times higher than in the gills, significantly enhancing ammonia nitrogen removal capacity. This is attributed to the liver's important role in detoxification in aquatic animals. GDH expression was initially inhibited at the onset of stress, followed by a gradual increase. This phenomenon relates to the high concentration of ammonia nitrogen in the water body. As stress time prolonged, the fish gradually adapted to the high ammonia nitrogen environment, accelerating ammonia nitrogen conversion and glutamate synthesis. Consequently, gills synthesize glutamine through GDH amination and transport it to the liver. GDH acts in combination with GS, playing a crucial role in reducing ammonia nitrogen toxicity and protecting aquatic organisms (Wang et al., 2014b).

To further clarify the causes of acute carbonate stress-induced injury in fish, the application of histomorphometric observations is an important indicator of the health status of the organism of aquatic animals affected by polluted environments. Gills are organs that directly interface with the aquatic environment, they serve as the primary site for gas exchange in fish. Here, blood acquires dissolved oxygen and ions such as

Statistics of GO Enrichment

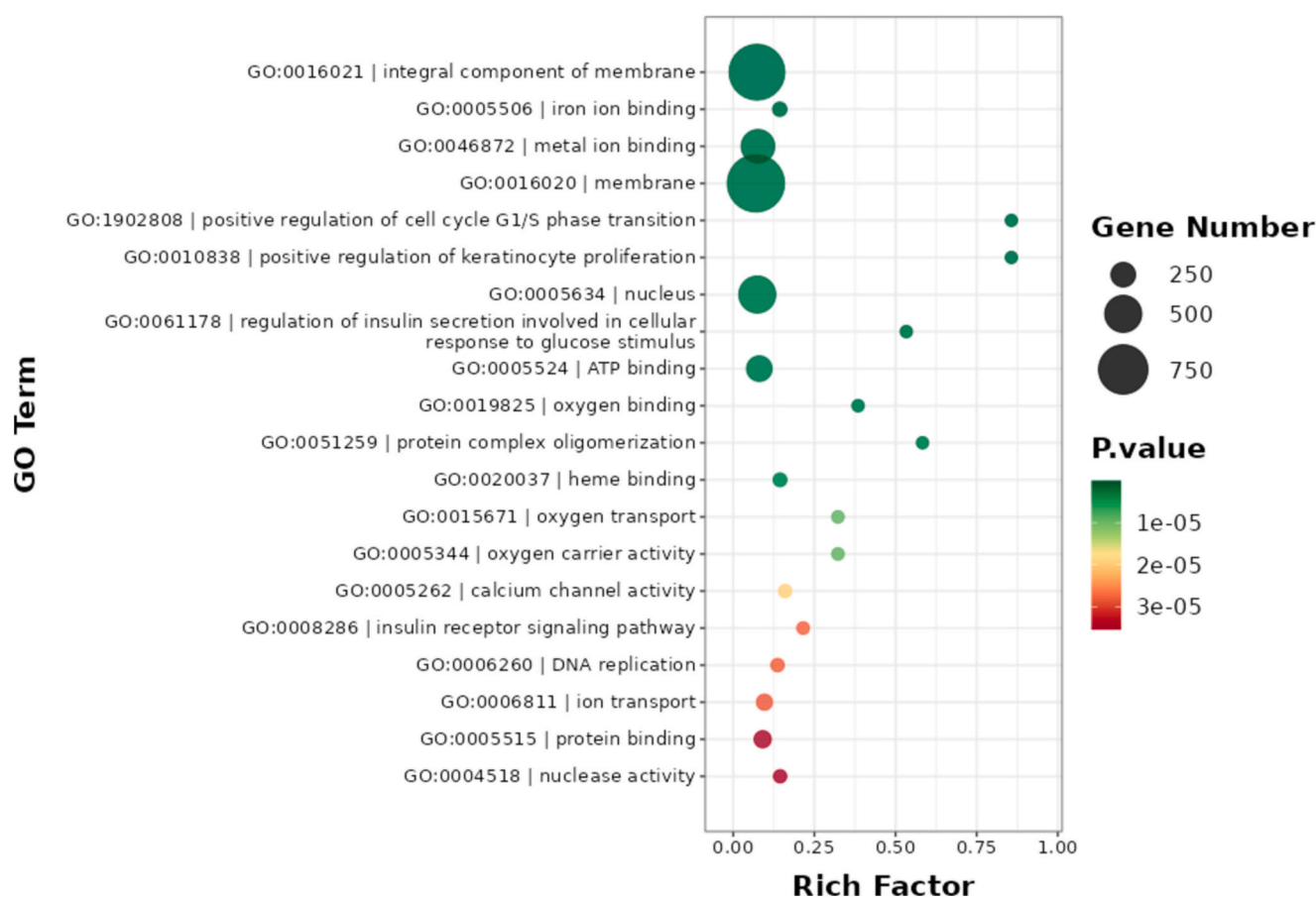


Fig. 14. Differential gene GO enrichment scatter plot. The horizontal axis represents the enrichment degree (Rich Factor), and the vertical axis represents the enriched GO Term. The size of the dots indicates the number of differential genes enriched in a GO term. Dot colors indicate different p-values. Rich factor represents the number of differential genes belonging to a GO term or the total number of genes belonging to this GO. The larger the Rich factor, the higher the GO enrichment.

HCO_3^- , CO_3^{2-} , and OH^- from saline water, which are then distributed to tissues throughout the body via the circulatory system (Alper et al., 2002; Kumar et al., 2020). Histological studies have shown that some fish species respond to adverse stress by altering gill respiratory function through tissue plasticity. The extent of these changes is generally less pronounced under long-term stress compared to short-term stress (Sinha et al., 2014).

Under hypoxic conditions, fish gradually lose ILCM through a combination of reduced mitotic rate and increased apoptotic rate, resulting in the emergence of a “growth layer” that increases in size and effectively increases gas exchange (Sollid and Nilsson, 2006). Similarly, fluctuations in temperature and salinity can induce analogous effects. For instance, the gill structure of blunt snout bream (*Megalobrama amblycephala*) undergoes remodeling in response to temperature changes, characterized by a decrease in ILCM height and an increase in mean lamellar area (Wu et al., 2017). In the present study, we also observed similar results that Hefang crucian carp were able to adapt to the respiratory impairment caused by acute carbonate alkalinity stress by gill remodeling to reduce the thickness of gill filaments and gill lamellae, to decrease the number of ILCMs as well as to increase the height of lamellae. This is because the fish body is in a highly permeable saline water environment, the oxygen consumption will increase, in order to improve the water oxygen exchange amount, the small gill plate shortening can make it better survival.

Under alkaline stress, endogenous ammonia excretion is blocked, and exogenous ammonia accumulates in the body, stimulating the

production of reactive oxygen species (ROS) in the liver and pancreas. This leads to oxidative damage, immunosuppression, and lipid peroxidation (Wilson and Taylor, 1992). The liver, a crucial hub for metabolism and immune defense, plays a vital role in energy storage and supply, helping to mitigate stress-induced damage (Wang et al., 2019). Excessive carbonate alkalinity affects liver energy expenditure, enzyme vitality, and tissue structure, potentially leading to severe consequences, including death. Serum biochemical indices, particularly aspartate aminotransferase (AST) and alanine aminotransferase (ALT), reflect impaired liver function. In adverse water conditions, increased blood ion concentration and liver cell membrane permeability result in the release of these transaminases into the blood (Sharf and Khan, 2022). The current study observed concentration-dependent increases in serum AST and ALT levels with carbonate exposure, indicating liver damage. This confirms the interaction between gill and liver biology under carbonate alkaline stress, reflecting the physiological changes in fish exposed to carbonate.

Previous studies have shown that salinity stress induces structural damage to fish liver tissue, including hepatocyte vacuolization, nuclear deviation, and nucleolysis, affecting normal physiological metabolism and health (McCormick, 2001; Tseng and Hwang, 2008). For example, chum salmon (*Oncorhynchus keta*) exposed to prolonged salinity stress exhibited localized liver necrosis (Liu et al., 2010), while Mozambique tilapia (*Oreochromis mossambicus*) showed hepatocyte degeneration and pigment deposition under acute salinity stress (Li, 2023). The present experiment observed similar liver histopathology. After 96 h of acute

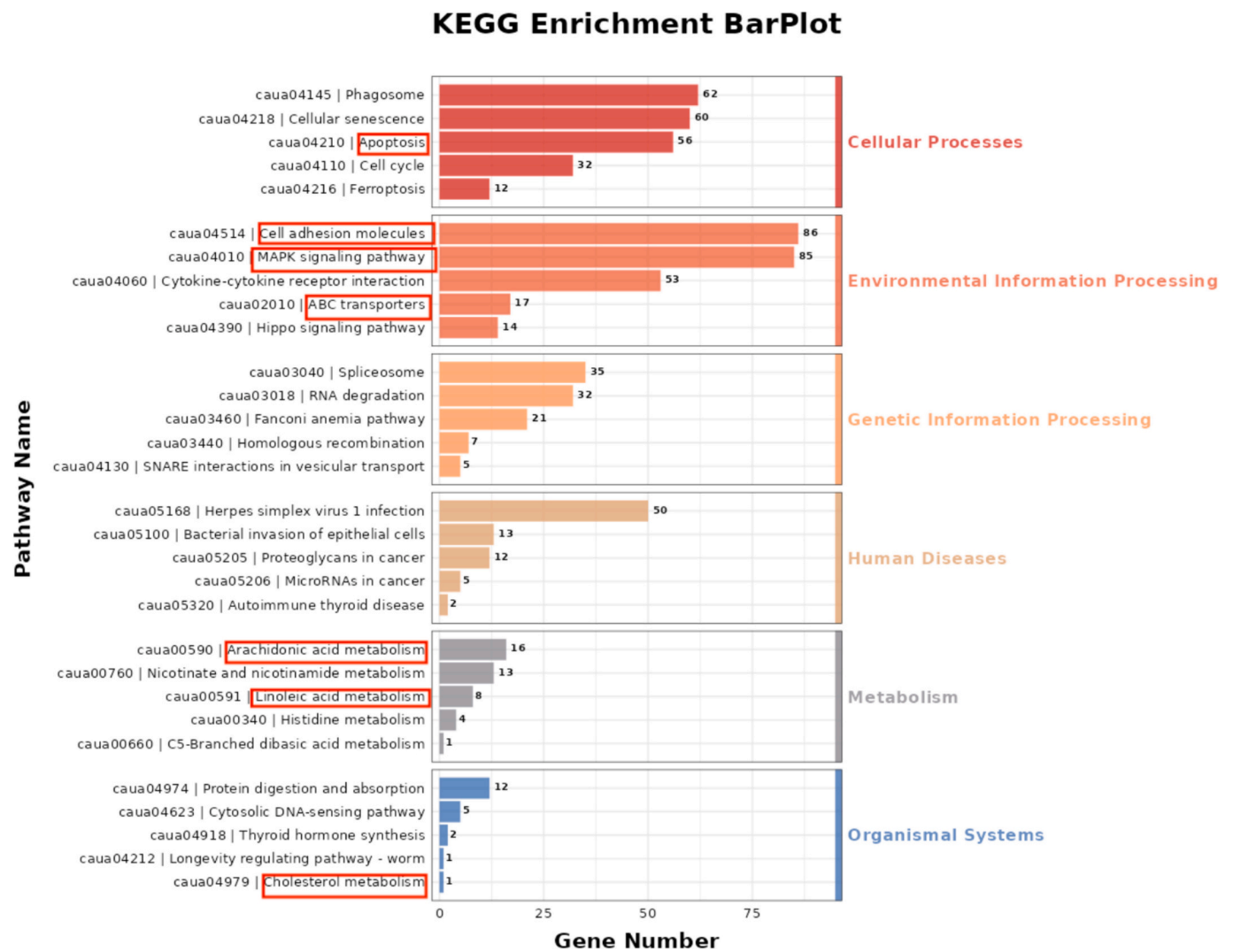


Fig. 15. KEGG enrichment classification bar chart of differential genes. From top to bottom, in descending order of number of genes.

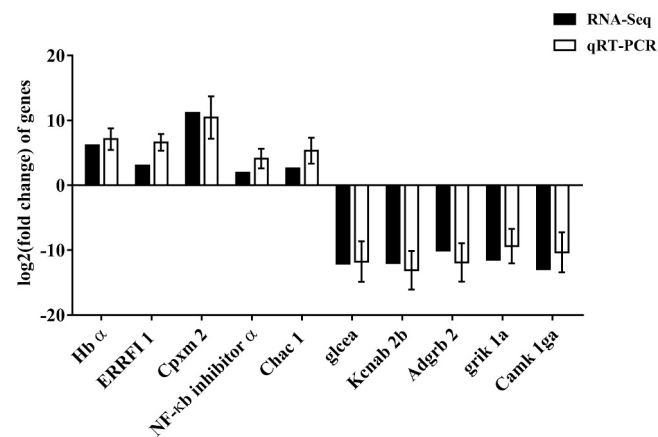


Fig. 16. Transcriptome results were verified by qRT-PCR.

carbonate stress, the low concentration (LC) group showed the appearance of sinusoidal channels, indicating slight liver tissue damage. As carbonate concentration increased, gill ion intake intensified, pushing the water environment beyond the fish's physiological regulation range and causing progressive structural and functional liver damage.

Fluctuations in the aquaculture environment compromise cell function, leading to ROS-induced lipid peroxidation (Lushchak, 2011). Fish activate antioxidant enzymes (SOD, CAT, and GSH-Px) to remove excessive ROS, reduce oxidative stress damage, and maintain normal bodily functions. SOD catalyzes the conversion of superoxide anions to hydrogen peroxide (H₂O₂) and molecular oxygen (O₂). Excessive H₂O₂ is then reduced to oxygen and water by CAT and GPH-Px, forming an effective protection system against ROS damage (Choi et al., 2008). MDA, as one of the markers of oxidative stress and an end product of membrane lipid peroxidation, also increases under stress conditions (Yang et al., 2019). With the extension of carbonate stress time, SOD, CAT, and GSH-Px levels increased first and then decreased. This is consistent with the trend of antioxidant enzyme activity in Sun et al. (Sun et al., 2012). In this study, the content change of MDA is exactly opposite to SOD and CAT, and the changes of the three can be used to evaluate the health of fish bodies under the alkalinity change. It indicates that salinity changes in the water column enhance the antioxidant function of the fish liver to scavenge excess oxidized free radicals in the body, as well as enhance the capacity of lipid peroxidation products, which leads to a decrease in MDA activity. The results indicate that at low carbonate stress concentrations, small amounts of ammonia and nitrogen entering through the gills significantly increased antioxidant enzyme activity. As stress concentration increased, the large amounts of oxygen free radicals produced required more antioxidant enzymes to

mitigate oxidative stress damage.

Elevated blood ammonia levels in fish trigger the production of immunosuppressive factors by peripheral lymphoid tissues, leading to decreased lymphocytes and phagocytes, suppressing the immune response (Barton and Iwama, 1991). This study used IgM, complement C3, complement C4, ACP, and AKP as indicators of immune response to harmful free radicals and phosphorylated products (Ayiku et al., 2020). Complement, mainly present in serum, is a crucial component of resistance to adverse environmental conditions. Environmental stimulation activates complement systems, increasing complement content for immune function (Gallani et al., 2019). IgM, the main fish immunoglobulin, recognizes and neutralizes invading pathogens in vitro (Cerezuela et al., 2012). The study observed that serum IgM, complement C3, and complement C4 activities initially decreased and then increased with stress duration. This aligns with findings by Qin et al., where IgM and C3 expression in darkbarbel catfish (*Pelteobagrus vachellii*) was suppressed after 96 h of ammonia nitrogen exposure, with reduced serum complement C3 content (Qin et al., 2017). AKP and ACP in small yellow fish (*Larimichthys polyactis*) liver showed distinct trends with increasing water column salinity (Wang et al., 2021), similar to the present study's results. ACP, involved in biophosphatase metabolism, protein hydrolysis, and calcium metabolism, is typically expressed at low levels under normal conditions but increases with organism damage. AKP, crucial in transport and absorption, plays a significant role in defense metabolism (Jiang et al., 2012). Under alkaline stress, Hefang crucian carp's immune response was initially suppressed. However, as stress continued, the fish increased antioxidant enzyme, AKP, and ACP activities through the immune system to cope with acute alkalinity stress, improving immune ability and slowing liver tissue damage.

The present study observed a trend of decreasing and then increasing MDA content in the liver with rising carbonate levels. This pattern indicates that carbonate exposure induces lipid peroxidation in the organism, with high lipid levels causing oxidative damage and leading to the release of large amounts of MDA from the liver into the serum. Blood lipids, as components of body lipids, reflect lipid metabolism in fish (Yan et al., 2021). Under stress, triglycerides play a crucial role in supporting the metabolic demands of parenchymal tissues and promoting tolerance, and together with total cholesterol, determine whether lipid accumulation occurs in the body (Liu et al., 2023). In this study, the levels of TG, T-CHO, LDL, and HDL in Hefang crucian carp declined as carbonate concentration increased. Furthermore, higher alkalinity intensified this decrease, suggesting enhanced lipid metabolism to meet the energy needs of the fish. This is similar to the results of Wang et al. who studied the effect of long-term carbonate stress on serum biochemical indices of hybrid sturgeon (*Huso dauricus*♀ × *Acipenser schrenckii*♂) (Wang et al., 2023). LDL and HDL are jointly involved in cholesterol transport. Meanwhile, accumulation of lipids in the body can lead to abnormal liver metabolism, steatosis, fatty liver and other diseases.

ACS, ACC, and FAS are key enzymes involved in liver lipogenesis. Under the experimental conditions, the activities of these enzymes increased with rising carbonate concentration. This indicates that carbonate treatment can induce abnormal lipid metabolism in the liver of Hefang crucian carp and lead to lipid accumulation in the hepatopancreas. Similar findings have been reported for ammonia-nitrogen stress, which can increase lipid synthesis levels, leading to lipid accumulation in the liver by increasing fatty acid expression (Fang et al., 2013).

On the other hand, β -oxidation is an important pathway for lipolysis metabolism in fish. PPAR α as a transcription factor can induce the expression of fatty acid oxidation-related genes, which affects their enzyme activities (Peng et al., 2014). For example, the β -oxidation rate-limiting enzyme CPT-1 exhibited a declining trend in content across all groups over time, followed by an eventual increase. This suggests that fish can enhance the gene expression of CPT-1 and its enzyme activity by activating PPAR α , thereby reducing lipid accumulation in vivo. This is consistent with the findings of Shi et al. (Shi et al., 2018). ACS, ACC, and FAS are key enzymes involved in liver lipogenesis. Under the

experimental conditions, the activities of these enzymes increased with rising carbonate concentration. This indicates that carbonate treatment can induce abnormal lipid metabolism in the liver of Hefang crucian carp and lead to lipid accumulation in the hepatopancreas. Similar findings have been reported for ammonia-nitrogen stress, which can increase lipid synthesis levels, leading to lipid accumulation in the liver by increasing fatty acid expression (Fang et al., 2013).

Under carbonate stress, serum glucose levels increased, indicating significant activation of the stress system in Hefang crucian carp. This activation stimulates carbohydrate metabolism, promotes glucose release and hepatic glycogenolysis, and provides energy for osmotic pressure homeostasis regulation (Nakano et al., 1998). Serum glucose levels were higher in the high concentration (HC) group compared to the low concentration (LC) and control (Con) groups, suggesting that Hefang crucian carp required more energy production to adapt to high alkalinity stress.

Glycolysis and gluconeogenesis are important pathways for glucose metabolism. Early studies confirmed that fish adapted to survive in acidic and alkaline water environments develop adaptive compensatory mechanisms, showing accelerated glycolytic pathways in alkaline water, thereby increasing resistance to pH stress (Bhaskar and Govindappa, 1986). HK, involved in the first step of the glycolytic reaction, phosphorylates glucose to glucose 6-phosphate, reflecting the fish's ability to utilize exogenous glucose (Qin et al., 2017). In this study, liver HK activity decreased with increasing carbonate concentration, suggesting liver inhibition of glucose phosphorylation under carbonate exposure.

PFK and PK, as key and rate-limiting enzymes in the glycolytic pathway, play a critical role in catalyzing the production of pyruvate from glycolysis (Ruan et al., 2009). The activities of PFK and PK in the liver of Hefang crucian carp from the carbonate alkalinity stress group were higher than those of the Con group within 96 h. The increase of their activities indicated that the body's glycolysis was enhanced, and the body's demand for energy was increased, therefore, the vitality generally reflects the size of the actual reaction rate of glycolysis.

Glycolysis, on the other hand, is a biochemical process that is the opposite of glycolysis to maintain blood glucose stability in the body. The process of gluconeogenesis in the hepatopancreas is able to utilize amino acids to convert them into energy required for some of the body's osmotic pressure regulation, in this process, endogenous ammonia is produced in the fish, and the free amino acids released by other tissues can be transported into the gills to participate in gluconeogenesis together with the liver (Mcnamara et al., 2004). One study found that relative mRNA levels of phosphoenolpyruvate carboxykinase (PEPCK), a key enzyme in gluconeogenesis, were significantly upregulated in Nile tilapia when PPAR α was activated (Ning et al., 2016). The current study's results were similar, demonstrating that the gluconeogenesis process was associated with PPAR α activation. At the onset of carbonate stress, glycolysis intensified while gluconeogenesis was inhibited. As stress prolonged, glucose in the body was consumed in large quantities, necessitating replenishment through gluconeogenesis. Eventually, these two processes self-regulated through body metabolism to reach a new energy balance (Hochachka and Lutz, 2001).

RNA-seq technology was employed to investigate gene expression in the liver tissues of Hefang crucian carp under different carbonate alkalinity conditions. This comprehensive analysis of pathways and genes related to alkalinity stress response provides valuable information for understanding the molecular biological mechanisms of Hefang crucian carp adaptation to alkaline environments.

Based on the statistical analysis of KEGG pathway abundance, numerous differentially expressed genes (DEGs) were identified in the liver of Hefang crucian carp, including those related to cell adhesion molecules, cytokine receptor interactions, thyroid hormone synthesis, and mitogen-activated protein kinase (MAPK). These DEGs were significantly induced by alkaline stress, similar to Zhou et al.'s findings in naked carp responding to salinity changes, where DEGs were mostly enriched in pathways associated with cell-cell adhesion (Zhou et al.,

2022). MAPK, a key component of the protein phosphorylation cascade, plays an important role in apoptosis by phosphorylating downstream transcription factors such as p53, activating transcription factor 2 (ATF2), and Myc-2-like transcriptional regulators. This phosphorylation regulates stress signaling and related gene expression (Cheng et al., 2017). Studies have shown that oxidative stress triggers an excessive inflammatory response, which in turn can induce intracellular ROS accumulation and apoptosis. Iron death, a major feature of a novel type of cell death, is associated with inflammation and oxidative stress (Long et al., 2023). The transcriptome KEGG enrichment analysis in this study revealed significant enrichment of apoptosis and iron death pathways, further suggesting that carbonate exposure induces oxidative stress and iron death, ultimately leading to apoptosis.

Additionally, a large number of genes involved in substance transport were affected. The expression of genes related to calcium and potassium ion channels, ATP-sensitive inwardly rectifying potassium channel 11-like, and sodium/potassium/calcium exchanger 4-like was down-regulated. This suggests that carbonate stress may affect the osmoregulatory capacity of gills by interfering with the transmembrane transport of these cations and ATPases, leading to increased cell membrane permeability and accumulation of exogenous ammonia-nitrogen in vivo, resulting in oxidative damage and lipid peroxidation in the liver. Concurrently, gene expression of several nutrient amino acids, sugar metabolism, and arachidonic acid transporter proteins was up-regulated. This indicates that carbonate stress can also affect the homeostasis of hepatic substance metabolism by interfering with the transport of these substances.

The pathological changes observed in gill and liver tissues in the current study are consistent with the regulation of metabolism, suggesting a link between these processes. Bu et al. (Bu et al., 2021) reported that multiple pathways associated with lipid metabolism, including fatty acid biosynthesis, PPAR signaling pathway, glycerophospholipid metabolism, pyruvate metabolism, and glutathione metabolism pathway, were heavily enriched in the livers of Mozambique tilapia subjected to prolonged salinity stress. Su et al. recently showed that salinity stress significantly altered lipid homeostasis in the gills of Mozambique tilapia, with pathways involved in arachidonic acid metabolism and linoleic acid metabolism enriched in all treatment groups (Su et al., 2023). The present study of liver transcriptomics produced similar results: gene expression of several nutritive amino acids, sugar metabolism, and arachidonic acid transport proteins were upregulated, and both linoleic acid metabolism and steroid biosynthesis pathways were significantly altered. These findings further support the notion that carbonate stress can affect the homeostasis of hepatic substance metabolism by interfering with the transport of these substances.

5. Conclusions

This study demonstrates that Hefang crucian carp (*Carassius auratus*) effectively adapt to carbonate stress through strategic utilization of carbohydrates for energy production and adaptive changes in their

osmoregulatory system. Excessive carbonate intake significantly alters blood serum physiological functions and liver metabolic processes. Transcriptomic analysis reveals that under carbonate stress, crucian carp primarily regulate their hepatic antioxidant system to combat oxidative stress, increase carbohydrate catabolism to meet energy demands, and modify lipid metabolism to assist in osmoregulation. These findings provide valuable insights into the physiological and molecular mechanisms underlying fish adaptation to alkaline environments, potentially informing strategies for aquaculture in saline-alkaline waters.

CRediT authorship contribution statement

Xinyu Lei: Writing – original draft, Validation, Methodology, Investigation, Formal analysis. Xue Cao: Writing – original draft, Methodology, Investigation, Formal analysis. Faye Zhang: Formal analysis. Qifang Lai: Formal analysis. Pengcheng Gao: Formal analysis. Yue-hong Li: Project administration, Funding acquisition, Conceptualization.

Ethics approval

The experiment was carried out in accordance with the research plan of the Institutional Animal Care and Use Committee of Jilin Agricultural University.

Consent to participate

All authors agree to participate in this study.

Consent to publication

All authors agree to participate in the publication of this article.

Code availability

Not applicable.

Funding

This research is supported by the China National Key Research and Development Plan Project (2023YFD2401003), the Jilin Province Young and Middle-aged Science and Technology Innovation and Entrepreneurship Outstanding Talent Project (20240601068RC) and the Jilin Province Science and Technology Development Plan Key research and Development Project (20230202076NC).

Declaration of competing interest

The authors have no relevant financial or non-financial interests to disclose.

Appendix A

A.1. Feed nutrient composition

Crude protein ≥	Coarse fibre ≤	Crude fat ≥	Crude ash ≤	Calcium	Total phosphorus ≥	Moisture ≤	Lysine ≥
32.00%	12.00%	4.00%	15.0%	0.50-2.00%	0.60%	12.5%	1.30

A.2. Kit information

Number	Name	Principal reagent	Wavelength (nm)	Company
1	Sodium (Na) Colorimetric Assay Kit	Chromogenic Agent, Enzyme Stock Solution, Enzyme Reagent, 10 mmol/L Standard	405	Elabscience
2	Potassium (K) Turbidimetric Assay Kit	Precipitant A, Precipitant B, Chromogenic Agent A, Chromogenic Agent B, 1 mmol/L Potassium Standard	450	Elabscience
3	NA+K+-ATPase assay kit	10mmol/L standard phosphorus reserve solution, Normal saline	636	Nanjing Jiancheng Bioengineering Institute
4	Blood ammonia assay kit	Protein precipitator I, Protein precipitator II, Color developer I, Color developer II, 7mmol/L standard reserve solution	630	Nanjing Jiancheng Bioengineering Institute
5	Urea Assay Kit	Basic sodium hypochlorite, BUN standard	640	Nanjing Jiancheng Bioengineering Institute
6	Glutamine synthetase assay kit	Extract	540	Nanjing Jiancheng Bioengineering Institute
7	Aspartate aminotransferase Assay Kit	Aspartate aminotransferase matrix solution, 4mol/L sodium hydroxide solution	510	Nanjing Jiancheng Bioengineering Institute
8	Alanine aminotransferase Assay Kit	Alanine aminotransferase matrix solution, matrix solution,	505	Nanjing Jiancheng Bioengineering Institute
9	Superoxide Dismutase (SOD) assay kit	Substrate reserve liquid, Buffer fluid	450	Nanjing Jiancheng Bioengineering Institute
10	CATase (CAT) assay kit	Substrate liquid, Developing powder	405	Nanjing Jiancheng Bioengineering Institute
11	Glutathione Peroxidase (GSH-PX) assay kit	1mmol/L GSH standard solution, Normal saline	412	Nanjing Jiancheng Bioengineering Institute
12	Malondialdehyde (MDA) assay kit	Glacial acetic acid, 10nmol/mL tetraethoxy propane	532	Nanjing Jiancheng Bioengineering Institute
13	Complement 3 Assay Kit	NaCl, NaN ₃ , Tris buffer, C3 antibody	340	Nanjing Jiancheng Bioengineering Institute
14	Complement C4 Test Kit	NaCl, NaN ₃ , Tris buffer, C4 antibody	340	Nanjing Jiancheng Bioengineering Institute
15	Acid phosphatase assay kit	Lye, 1.1mg/mL phenol standard reserve solution,	520	Nanjing Jiancheng Bioengineering Institute
16	Alkaline phosphatase assay kit	Buffer, Matrix fluid, Color developing agent	520	Nanjing Jiancheng Bioengineering Institute
17	Triglyceride assay kit	ATP, Glycerol kinase, Peroxidase	500	Nanjing Jiancheng Bioengineering Institute
18	Total cholesterol assay kit	4-AAP, Bovine serum albumin, Cholesterol esterase	500	Nanjing Jiancheng Bioengineering Institute
19	Low-density lipoprotein cholesterol assay kit	Calibration product, Phosphate buffer	600	Nanjing Jiancheng Bioengineering Institute
20	High-density lipoprotein cholesterol assay kit	Calibration product, Phosphate buffer	600	Nanjing Jiancheng Bioengineering Institute
21	Glucose Assay Kit	Magnesium chloride, DHBS, Glucose oxidase	505	Nanjing Jiancheng Bioengineering Institute
22	Liver glycogen assay kit	Alkaline solution, 1mg/mL glucose standard storage solution	620	Nanjing Jiancheng Bioengineering Institute
23	Glutamate dehydrogenase Assay Kit	Extraction Solution, Buffer Solution, Substrate, Substrate	450	Elabscience
24	Immunoglobulin M Assay Kit	Specialty protein calibrators	340	Nanjing Jiancheng Bioengineering Institute
25	Acetyl-CoA Carboxylase (ACC) Activity Assay Kit	500 μmol/L Standard Solution, Acid Solution, Chromogenic Agent	660	Elabscience
26	Fish CPT1 ELISA Kit	Standard goods, Sample diluent, Detection of antibody HRP, Stop solution	450	Renjie Biology
27	Acetyl-CoA synthetase ELISA kit	Reagent one, two, three, four	340	Shanghai Enzyme-linked Biotechnology Co., Ltd
28	Hepatic Lipas assay kit	Chloroform, 2mmol/L palmitic acid standard storage solution	550	Nanjing Jiancheng Bioengineering Institute
29	Lipoprotein lipase assay kit	Chloroform, 2mmol/L palmitic acid standard storage solution	550	Nanjing Jiancheng Bioengineering Institute
30	Hexokinase assay kit	Reagent one, two, three, four	340	Nanjing Jiancheng Bioengineering Institute
31	Phosphofructokinase (PFK) test kit	Reagent one, two, three, four	340	Nanjing Jiancheng Bioengineering Institute
32	Pyruvate kinase assay kit	Reagent one, two, three, four	340	Nanjing Jiancheng Bioengineering Institute
33	Phosphoenolpyruvate carboxylase kinase Kit	Reagent one, two, three, four	340	Nanjing Jiancheng Bioengineering Institute

Data availability

Data will be made available on request.

References

- Alper, S.L., Darman, R.B., Chernova, M.N., 2002. The AE gene family of Cl/HCO₃-exchangers. *J. Nephrol.* 5, S41.
- Alves, R.N., Cordeiro, O., Silva, Tomé S., 2010. Metabolic molecular indicators of chronic stress in gilthead seabream (*Sparus aurata*) using comparative proteomics. *Aquaculture* 299 (1–4), 57–66.

- Anderson, P.M., Broderius, M.A., Fong, K.C., 2002. Glutamine synthetase expression in liver, muscle, stomach and intestine of *Bostrichthys sinensis* in response to exposure to a high exogenous ammonia concentration. *J. Exp. Biol.* 14, 2053–2065.
- Ayiku, S., Jianfei, S., Tan, B.P., 2020. Effects of dietary yeast culture on shrimp growth, immune response, intestinal health and disease resistance against *Vibrio harveyi*. *Fish Shellfish Immunol.* 102, 286–295.
- Barton, B.A., Iwama, G.K., 1991. Physiological changes in fish from stress in aquaculture with emphasis on the response and effects of corticosteroids. *Annu. Rev. Fish Dis.* 1, 3–26.
- Bhaskar, M., Govindappa, S., 1986. Effect of environmental acidity and alkalinity on the physiology of *Tilapia mossambica* during acclimation. *Biochem. Syst. Ecol.* 14 (4), 439–443.
- Bu, X.J., Zhu, J.H., Liu, S.B., 2021. Growth, osmotic response and transcriptome response of the euryhaline teleost, *Oreochromis mossambicus* fed different myo-inositol levels under long-term salinity stress. *Aquaculture* 534, 736294.
- Cerezuela, R., Guardiola, A.F., Meseguer, J., 2012. Increases in immune parameters by inulin and *Bacillus subtilis* dietary administration to gilthead seabream (*Sparus aurata* L.) did not correlate with disease resistance to *Photobacterium damsela*. *Fish Shellfish Immunol.* 32 (6), 1032–1040.
- Cheng, C.H., Ye, C.X., Guo, Z.X., 2017. Immune and physiological responses of pufferfish (*Takifugu obscurus*) under cold stress. *Fish Shellfish Immunol.* 64, 137–145.
- Choi, C.Y., An, K.W., An, M.I., 2008. Molecular characterization and mRNA expression of glutathione peroxidase and glutathione S-transferase during osmotic stress in olive flounder (*Paralichthys olivaceus*). *Comp. Biochem. Physiol. A Mol. Integr. Physiol.* 149 (3), 330–337.
- Chun, Y.S., Cheng, S.H., Zhu, M.Y., 2020. Exploring the metabolic biomarkers and pathway changes in crucian under carbonate alkalinity exposure using high-throughput metabolomics analysis based on UPLC-ESI-QTOF-MS. *RSC Adv.* 10 (3), 1552–1571.
- Cui, Q., Qiu, L., Yang, X., 2019. Transcriptome profiling of the low-salinity stress responses in the gills of the juvenile *Pseudopleuronectes yokohamae*. *Comp. Biochem. Physiol. Part D Genomics Proteomics* 32, 100612.
- Evans, D.H., 2005. The multifunctional fish gill: dominant site of gas exchange, osmoregulation, acid-base regulation, and excretion of nitrogenous waste. *Physiol. Rev.* 85 (1), 97–177.
- Fan, Z., Wu, D., Zhang, Y.Y., 2021. Carbonate alkalinity and dietary protein levels affected growth performance, intestinal immune responses and intestinal microflora in Songpu mirror carp (*Cyprinus carpio* Songpu). *Aquaculture* 3, 737135.
- Fang, D.L., Wan, Y., Shen, W., 2013. Endoplasmic reticulum stress leads to lipid accumulation through upregulation of SREBP-1c in normal hepatic and hepatoma cells. *Mol. Cell. Biochem.* 381 (1–2), 127–137.
- Fang, L., Bai, X.L., Liang, X.F., et al., 2017. Ammonia nitrogen excretion in mandarin fish (*Siniperca chuatsi*) and grass carp (*Ctenopharyngodon idellus*) fed practical diets: the effects of water temperature. *Aquac. Res.* 48 (3), 836–843.
- Feng, D.X., Yang, Z.G., Li, M., 2023. Dietary N-carbamylglutamate supplementation improves ammonia tolerance of juvenile yellow catfish *Pelteobagrus fulvidraco*. *Front. Physiol.* 14, 1191468.
- Franklin, C.E., 2010. Surface ultrastructural changes in the gills of sockeye salmon (teleostei: *Oncorhynchus nerka*) during seawater transfer: comparison of successful and unsuccessful seawater adaptation. *J. Morphol.* 206 (1), 13–23.
- Gallani, S.U., Vallado, G.M.R., Kotzent, S., 2019. Dietary intake of *Rubrivivax* gelatinosus biomass enhances phagocytic cells in tropical fish *Piaractus mesopotamicus* infected with *Aeromonas hydrophila*. *Aquac. Int.* 27 (3), 711–720.
- Gao, N., Zhu, L., Guo, Z., 2017. Effects of chronic ammonia exposure on ammonia metabolism and excretion in marine medaka *Oryzias latipes*. *Fish Shellfish Immunol.* 65, 226–234.
- Guo, W.F., Yao, Z.L., Lai, Q.F., et al., 2014. Differential gene expression of *Gymnocypris przewalskii* under saline-alkali stress. *Mar. Fish.* 34 (2), 137–144.
- Hegazi, M.M., Attia, Z.I., Hegazi, M.A.M., 2010. Metabolic consequences of chronic sublethal ammonia exposure at cellular and subcellular levels in Nile tilapia brain. *Aquaculture* 299 (1–4), 149–156.
- Hochachka, P.W., Lutz, P.L., 2001. Mechanism, origin, and evolution of anoxia tolerance in animals. *Comp. Biochem. Physiol. B Biochem. Mol. Biol.* 130 (4), 435–459.
- Jiang, H., Yang, H., Kong, X., 2012. Response of acid and alkaline phosphatase activities to copper exposure and recovery in freshwater fish *Carassius auratus gibelio* var. *Life Sci.* 9 (3), 233–245.
- Jiang, W., Tian, X., Fang, Z., 2019. Metabolic responses in the gills of tongue sole (*Cynoglossus semilaevis*) exposed to salinity stress using NMR-based metabolomics. *Sci. Total Environ.* 653, 465–474.
- Jin, X., Meng, L., Qi, Z., 2024. Effects of dietary selenium deficiency and supplementation on liver in grazing sheep: insights from transcriptomic and metabolomic analysis. *Front. Vet. Sci.* 11, 1358975.
- Kultz, D., 2015. Physiological mechanisms used by fish to cope with salinity stress. *J. Exp. Biol.* 218, 1907–1914.
- Kumar, N., Chandan, N.K., Wakchaure, C.G., 2020. Synergistic effect of zinc nanoparticles and temperature on acute toxicity with response to biochemical markers and histopathological attributes in fish. *Comp. Biochem. Physiol. C Toxicol. Pharmacol.* 229, 108678.
- Lei, Y.X., Cao, X., Sun, J., 2024. Effect of different concentrations of carbonate on growth performance, intestinal health and hepatic lipid metabolism of Crucian carp. *Aquaculture* 589, 740990.
- Li, H., Zhang, M., Jiang, H., 2021. Arginase plays an important role in ammonia detoxification of yellow catfish *Pelteobagrus fulvidraco*. *Fish Shellfish Immunol.* 115, 171–178.
- Li, P., Jia, L., Chen, Q., 2024. Adaptive evaluation for agricultural sustainability of different fertilizer management options for a green manure-maize rotation system: impacts on crop yield, soil biochemical properties and organic carbon fractions. *Sci. Total Environ.* 15, 908.
- Li, Y.Y., 2023. Tissue Structure, Physiological and Biochemical Response and Transcriptome Analysis of *Tilapia Mozambican* to Salt-alkali Stress. Shanghai ocean university (in Chinese).
- Lin, T.T., Zhou, K., Hao, Z.R., 2015. Carbonic anhydrase 2-like and Na⁺-K⁺-ATPase a gene expression in medaka (*Oryzias latipes*) under carbonate alkalinity stress. *Fish Physiol. Biochem.* 41 (6), 1491–1500.
- Listiyani, Shobirin, Novianti, 2018. Effect of Cd on serum osmolality, ion levels and hematological parameters of tilapia (*Oreochromis niloticus*) at different salinity levels. *Zool. Ecol.* 28 (3), 205–212.
- Liu, D., Zhang, Z., Song, Y., 2023. Effects of salinity on growth, physiology, biochemistry and gut microbiota of juvenile grass carp (*Ctenopharyngodon idella*). *Aquat. Toxicol.* 258, 106482.
- Liu, J.Y., Yao, Z.L., Lai, Q.F., et al., 2012. Effects of salt and alkali stress on oxygen consumption rate, plasma osmotic concentration and ion concentration of *Gymnocypris Qinghai Lake*. *Chin. J. Ecol.* 31 (3), 664–669 (in Chinese).
- Liu, W., Zhi, B.J., Zhan, P.R., 2010. Effects of salinity on blood biochemical indices and liver tissue of juvenile salmon. *Chin. J. Appl. Ecol.* 21 (9), 2411–2417 (in Chinese).
- Long, X., He, K., Zhang, M., 2023. Temporal correlations of ferroptosis, inflammation and oxidative stress under acute ammonia exposure in brain tissue of yellow catfish (*Pelteobagrus fulvidraco*). *Comp. Biochem. Physiol. Toxicol. Pharmacol.* 271, 109693.
- Longwu, G., Guangxiang, T., Haifeng, J., 2016. Effect of salinity and alkalinity on *Luciobarbus capito* gill Na⁺/K⁺-ATPase enzyme activity, plasma ion concentration, and osmotic pressure. *Biomed. Res. Int.* 2016, 1–7.
- Lu, Y.T., Zhang, Y.L., Zhang, P.J., 2022. Effects of dietary supplementation with *Bacillus subtilis* on immune, antioxidant, and histopathological parameters of *Carassius auratus gibelio* juveniles exposed to acute saline-alkaline conditions. *Aquac. Int.* 30 (5), 2295–2310.
- Lushchak, V.I., 2011. Environmentally induced oxidative stress in aquatic animals. *Aquat. Toxicol.* 101, 13–30.
- McCormick, S.D., 2001. Endocrine control of osmoregulation in teleost fish. *Am. Zool.* 41 (4), 781–794.
- Mcnamara, J.C., Rosa, José C., Greene, L.J., 2004. Free amino acid pools as effectors of osmotic adjustment in different tissues of the freshwater shrimp *Macrobrachium olfersii* (Crustacea, Decapoda) during long-term salinity acclimation. *Mar. Freshw. Behav. Physiol.* 37 (3), 193–208.
- Mylonas, C.C., Pavlidis, M., Papandroulakis, N., 2009. Growth performance and osmoregulation in the shi drum (*Umbrina cirrosa*) adapted to different environmental salinities. *Aquaculture* 287, 203–210.
- Nakano, K., Tagawa, M., Takemura, A., 1998. Temporal changes in liver carbohydrate metabolism associated with seawater transfer in *Oreochromis mossambicus*. *Comp. Biochem. Physiol. B.* 119 (4), 721–728.
- Nguyen, P., Leray, V., Diez, M., 2008. Liver lipid metabolism. *J. Anim. Physiol. Anim. Nutr.* 92 (3), 272–283.
- Ning, L.J., He, A.Y., Li, J.M., 2016. Mechanisms and metabolic regulation of PPAR α activation in Nile tilapia (*Oreochromis niloticus*). *Biochim. Biophys. Acta Mol. Cell Biol. Lipids* 1861 (9), 1036–1048.
- Nong, X., 2023. Saline-alkali land fishery opens up a new track of aquaculture. *Rural New Technol.* 11, 4–7.
- Peh, W.Y.X., Chew, S.F., Ching, B.Y., et al., 2010. Roles of intestinal glutamatedehydrogenase and glutamine synthetase in environmental ammoniatedetoxification in the euryhaline four-eyed sleeper, *Bostrichthys sinensis*. *Aquat. Toxicol.* 98 (1), 91–98.
- Peng, M., Xu, W., Mai, K., 2014. Growth performance, lipid deposition and hepatic lipid metabolism related gene expression in juvenile turbot (*scophthalmus maximus* L.) fed diets with various fish oil substitution levels by soybean oil. *Aquaculture* 433, 442–449.
- Peng, R.B., Le, K.X., Wang, P.S., 2017. Detoxification pathways in response to environmental ammonia exposure of the cuttlefish, *Sepia pharaonis*: glutamine and urea formation. *J. World Aquacult. Soc.* 48 (2), 342–352.
- Perry, S.F., Gilmour, K.M., 2006. Acid-base balance and CO₂ excretion in fish: unanswered questions and emerging models. *Respir. Physiol. Neurobiol.* 154 (1–2), 199–215.
- Qin, C., Shao, T., Wang, Y., 2017. Effect of ammonia-N on histology and expression of immunoglobulin M and component C3 in the spleen and head kidney of *Pelteobagrus vachellii*. *Aquac. Res.* 8, 16–22.
- Ruan, K., Duan, J.B., Bai, F.W., 2009. Function of *Dunaliella salina* (*Dunaliellaceae*) enolase and its expression during stress. *Eur. J. Phycol.* 44 (2), 207–214.
- Schreck, C.B., Contreras-Sanchez, W., Fitzpatrick, M.S., 2011. Effects of stress on fish reproduction, gamete quality, and progeny. *Aquaculture* 197, 3–24.
- Sharf, Y., Khan, M.A., 2022. Dietary histidine requirement for fingerlings of *Channa punctatus* (Bloch). *Aquaculture* 561, 738732.
- Shi, X.C., Jin, A., Sun, J., 2018. The protein-sparing effect of α -lipoic acid in juvenile grass carp, *Ctenopharyngodon idellus*: effects on lipolysis, fatty acid β -oxidation and protein synthesis. *Br. J. Nutr.* 120 (9), 977–987.
- Sinha, A.K., Matey, V., Giblen, T., 2014. Gill remodeling in three freshwater teleosts in response to high environmental ammonia. *Aquat. Toxicol.* 155, 166–180.
- Sollid, J., Nilsson, G.E., 2006. Plasticity of respiratory structures: adaptive remodeling of fish gills induced by ambient oxygen and temperature. *Respir. Physiol. Neurobiol.* 154 (1–2), 241–251.
- Su, H., Ma, D.M., Fan, J.J., 2023. Metabolism response mechanism in the gill of *Oreochromis mossambicus* under salinity, alkalinity and saline-alkalinity stresses. *Ecotoxicol. Environ. Saf.* 251, 114523.

- Sun, H., Lue, K., Minter, E.J.A., 2012. Combined effects of ammonia and microcystin on survival, growth, antioxidant responses, and lipid peroxidation of bighead carp hypophthalmichthys nobilis larvae. *J. Hazard. Mater.* 30, 213–219.
- Sun, Y.C., Wu, S., Du, N.N., 2018. High-throughput metabolomics enables metabolite biomarkers and metabolic mechanism discovery of fish in response to alkalinity stress. *RSC Adv.* 8 (27), 14983–14990.
- Tseng, Y.C., Hwang, P.P., 2008. Some insights into energy metabolism for osmoregulation in fish. *Comp. Biochem. Physiol. Toxicol. Pharmacol.* 148 (4), 419–429.
- Wang, G., Pan, L., Ding, Y., 2014a. Defensive strategies in response to environmental ammonia exposure of the sea cucumber *Apostichopus japonicus*: glutamine and urea formation. *Aquaculture* 432, 278–285.
- Wang, M.J., Chu, T.Q., Liu, F., 2021. Effects of salinity stress on the viability of antioxidant enzymes, nonspecific immune enzymes and Na⁺/K⁺-ATP enzymes in small yellow fish (*Larimichthys polyactis*). *J. Oceanogr.* 43 (2), 59–66 (in Chinese).
- Wang, N., Yang, H., Feng, C.J., 2023. Effects of carbonate alkalinity on growth and serum biochemical indices of 3-month-old hybrid *Acipenser* (*Huso dauricus* × *Acipenser schrenckii*). *J. Shanghai Ocean Univ.* 32 (1), 98–107 (in Chinese).
- Wang, X.W., Zhang, R., Zhu, J.Y., 2019. Effects of acute heat stress on liver function indexes and liver heat shock protein expression of Siberian Sturgeon. *J. Sichuan Agric. Univ.* 37 (1), 122–128 (in Chinese).
- Wang, Y., Ye, Q.Q., Liu, C.X., 2014b. Involvement of Na/K-ATPase in hydrogen peroxide-induced activation of the Src/ERK pathway in LLC-PK1 cells. *Free Radic. Biol. Med.* 71, 415–426.
- Wang, Y.S., Gonzalez, R.J., Patrick, M.L., 2003. Unusual physiology of scale-less carp, *Gymnocypris przewalskii*, in Lake Qinghai: a high altitude alkaline saline lake. *Comp. Biochem. Physiol. A Mol. Integr. Physiol.* 134 (2), 409–421.
- Wilkie, P.M., 1997. Mechanisms of Ammonia excretion across fish gills. *Comp. Biochem. Physiol. A Physiol.* 118 (1), 39–50.
- Wilson, R.W., Taylor, E.W., 1992. Transbranchial ammonia gradients and acid-base responses to high external ammonia concentration in rainbow trout (*Oncorhynchus mykiss*) acclimated to different salinities. *J. Exp. Biol.* 166 (1), 95–112.
- Wu, C., Liu, Z., Li, F., 2017. Gill remodeling in response to hypoxia and temperature occurs in the hypoxia sensitive blunt snout bream (*Megalobrama amblycephala*). *Aquaculture* 479, 479–486.
- Yan, X., Dong, X., Tan, B., 2021. Effects of alternative dietary oils on lipid metabolism and related gene expression in hybrid grouper (♀ *Epinephelus fuscoguttatus* × ♂ *E. lanceolatus*). *Aquac. Nutr.* 27 (1), 105–115.
- Yang, Q., Liang, H., Mokrani, A., 2019. Dietary histidine affects intestinal antioxidant enzyme activities, antioxidant gene expressions and inflammatory factors in juvenile blunt snout bream (*Megalobrama amblycephala*). *Aquac. Nutr.* 25 (1), 249–259.
- Yonghang, Z., Haishen, W., Yang, L., 2023. Gill histological and transcriptomic analysis provides insights into the response of spotted sea bass (*Lateolabrax maculatus*) to alkalinity stress. *Aquaculture* 563 (1), 738945.
- Zhao, Y., Wu, J.W., Wang, Y., 2016. Role of miR-21 in alkalinity stress tolerance in tilapia. *Biochem. Biophys. Res. Commun.* 471 (1), 26–33.
- Zhou, B.Z., Qi, D.L., Liu, S.J., 2022. Physiological, morphological and transcriptomic responses of Tibetan naked carps (*Gymnocypris przewalskii*) to salinity variations. *Comp. Biochem. Physiol. Part D Genomics Proteomics* 42, 100982.
- Zhu, R., 2023. Effects of the Soy Antigen Glycinin on the Growth, Antioxidant Capacity, and Intestinal Tissue of *Phoxinus phalloides*. Jilin Agricultural University.